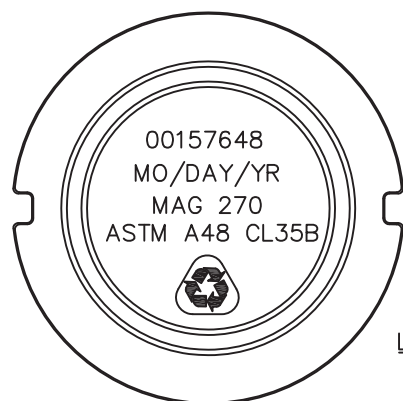
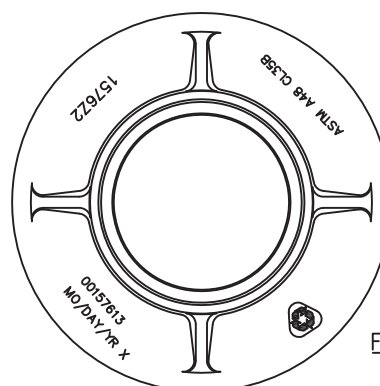
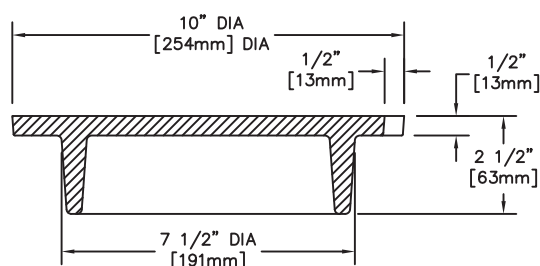
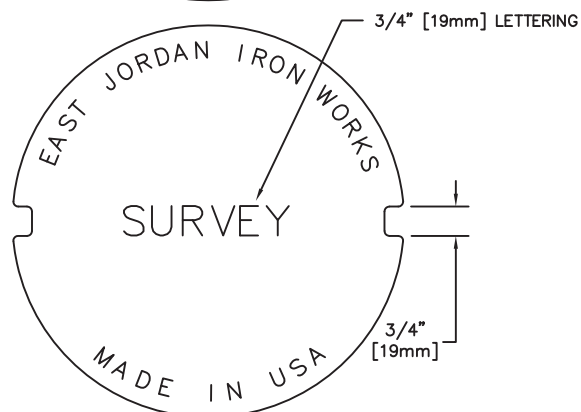




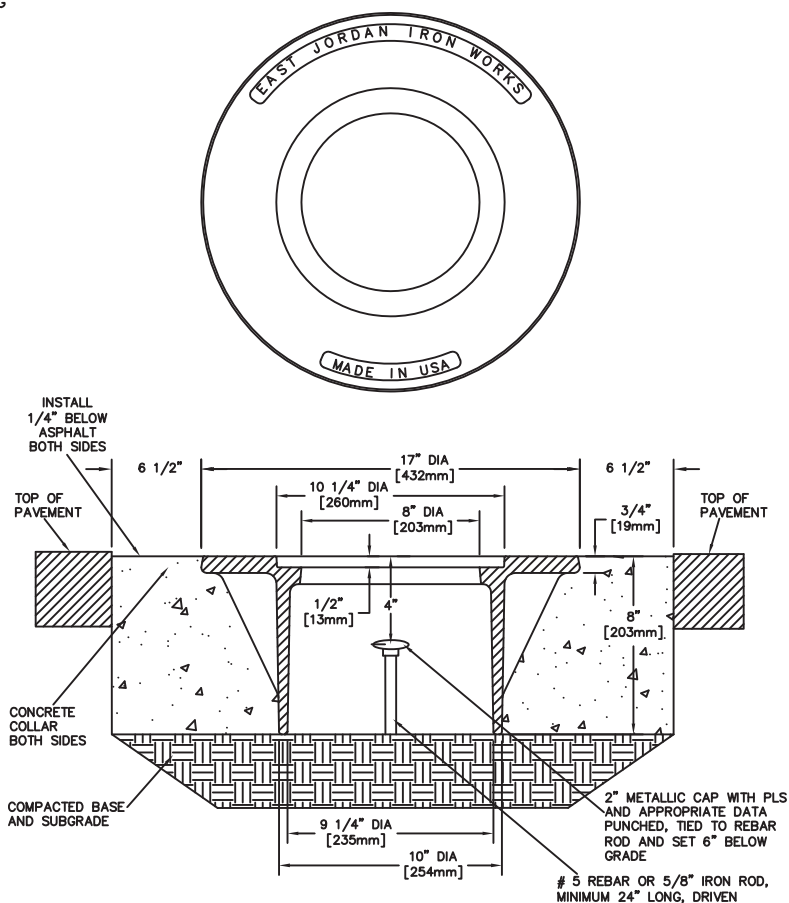
LID SECTION



FRAME SECTION



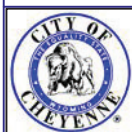
LID DETAIL



FRAME DETAIL

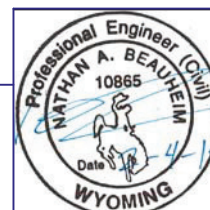
NOTES:

1. ADJUST MONUMENT BOX UPWARD OR DOWNWARD AS REQUIRED. FINAL ADJUSTMENT SHALL BE MADE AFTER PAVING AND BEFORE SEALING.
2. WHEN CONCRETE COLLAR IS PLACED ADJACENT TO CONCRETE PAVEMENT, COLLAR SHALL MATCH ADJACENT CONCRETE DEPTH WITH A MINIMUM OF 8" DEPTH.



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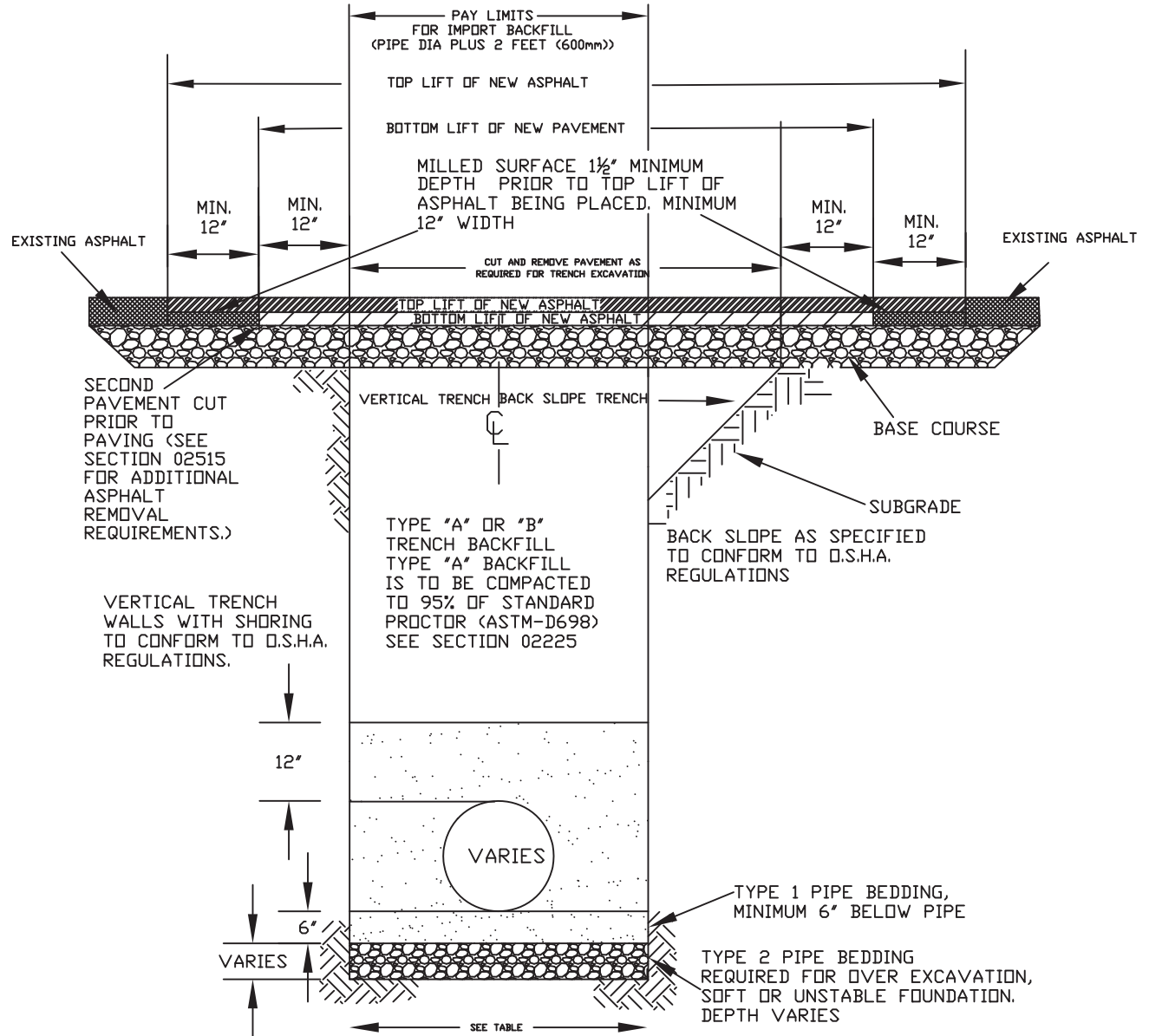
MONUMENT BOX
SETTING AND ADJUSTMENT DETAILS
(OPTION UPON APPROVAL)



NEW 2014

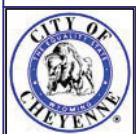
STANDARD
DRAWING NO.
01330-02

NOT TO SCALE



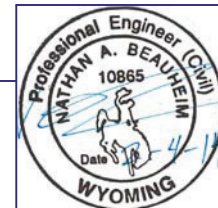
NOTES:

1. MATERIAL COMPACTION SHALL CONFORM TO SECTIONS 02220, 02225, 02226, 02231, AND 02512.
2. WHEN ANY PORTION OF THE PIPE HAS LESS THAN 18" COVER TO FINISH GRADE OR THE TRENCH IS LESS THAN 30" IN WIDTH, THE TRENCH SHALL BE BACKFILLED WITH CITY MIX 3 SLURRY.
3. DEPTH OF BITUMINOUS PATCH SHALL BE A MINIMUM OF 4" OR MATCH EXISTING UP TO 6" ON ALL STREETS DESIGNATED AS LOCAL STREETS OR ALLEYS. PATCH DEPTH OF 6" OR MATCH EXISTING UP TO 12" ARE REQUIRED ON ALL COLLECTORS AND ARTERIALS.
4. EXISTING PAVEMENT SHALL BE CUT BACK A MINIMUM OF 12" BEYOND ANY DISTURBED BASE COURSE, SUBGRADE OR BROKEN ASPHALT. IN ADDITION, THE ADJACENT PAVEMENTS SHALL BE MILLED A DEPTH OF $1\frac{1}{2}$ " NO LESS THAN ONE FOOT WIDE AND THE TOP LIFT INLAID. SEE SECTION 02515 FOR ADDITIONAL ASPHALT REMOVAL AND PATCHING REQUIREMENTS.



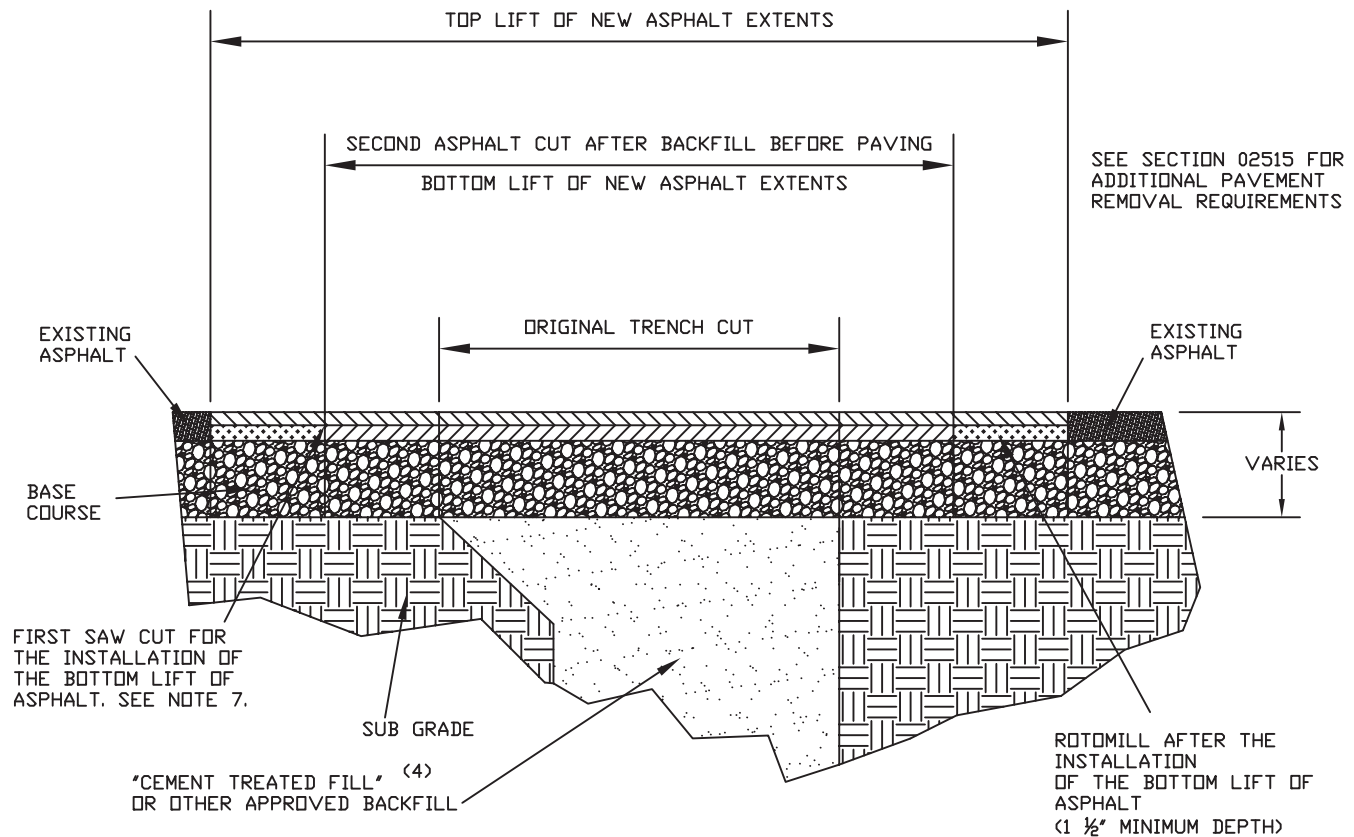
CITY OF CHEYENNE
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STANDARD DRAWING

TYPICAL TRENCH DETAIL
FOR STORM SEWER, WATER AND
SANITARY SEWER PIPE



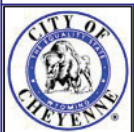
REVISED
2014

STANDARD
DRAWING NO.
02220-01



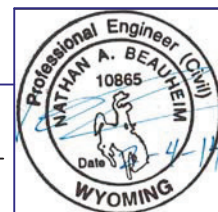
NOTES:

1. MATERIAL COMPACTION SHALL CONFORM TO SECTIONS 02220, 02225, 02226, 02231, AND 02512.
2. WHEN ANY PORTION OF THE PIPE HAS LESS THAN 18" OF COVER TO FINISH GRADE OR THE TRENCH IS LESS THAN 30" IN WIDTH THE TRENCH SHALL BE BACKFILLED WITH CITY MIX 3 SLURRY TO TOP OF SUBGRADE.
3. DEPTH OF BITUMINOUS PATCH SHALL BE A MINIMUM OF 4" OR MATCH EXISTING UP TO 6" ON ALL STREETS DESIGNATED AS LOCAL OR ALLEYS. PATCH DEPTH OF 6" OR MATCH EXISTING UP TO 12" ARE REQUIRED ON ALL STREETS DESIGNATED AS COLLECTOR OR ARTERIAL.
4. EXISTING PAVEMENT SHALL BE CUT BACK A MINIMUM OF 12" BEYOND ANY DISTURBED BASE, SUBGRADE OR ASPHALT. IN ADDITION, THE ADJACENT PAVEMENTS SHALL BE MILLED A DEPTH OF 1 1/2" NO LESS THAN ONE FOOT WIDE AND THE TOP LIFT OF ASPHALT INLAID. SEE SECTION 02515 FOR ADDITIONAL ASPHALT REMOVAL AND PATCHING REQUIREMENTS.



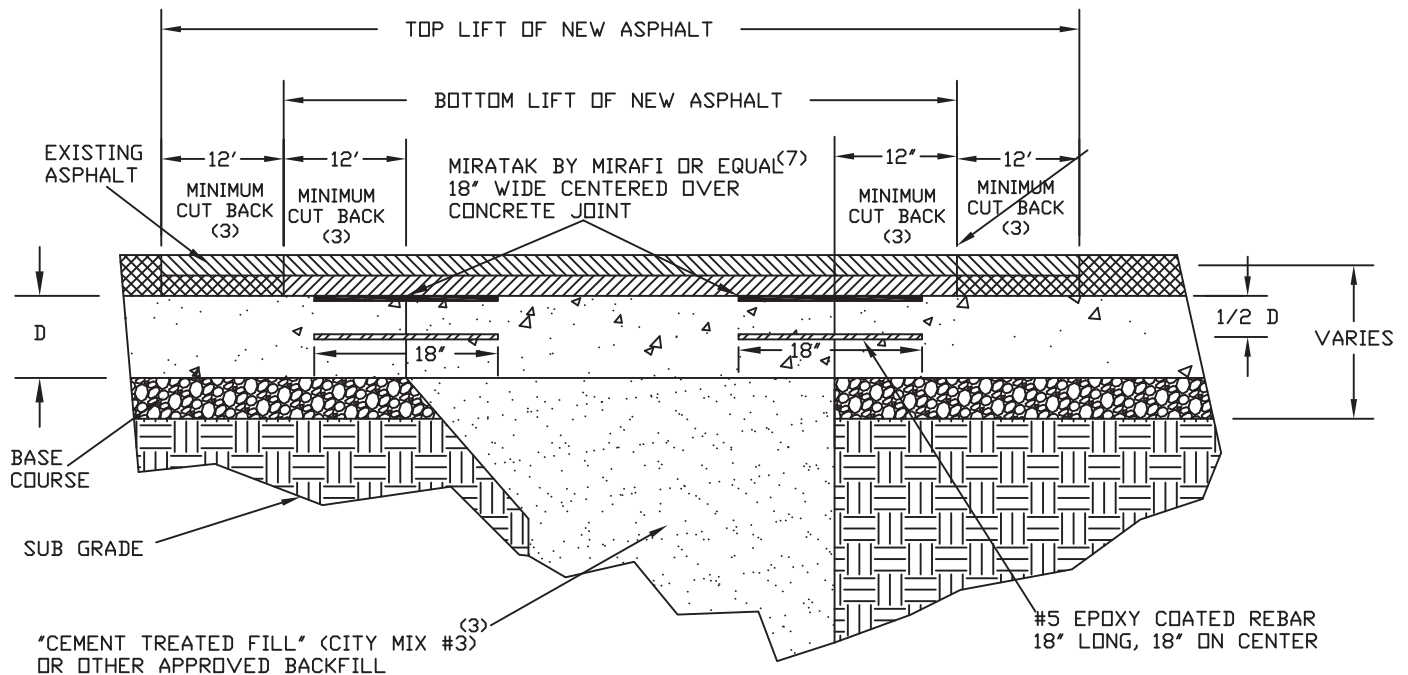
CITY OF CHEYENNE
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STREET CUT REPAIRS
REPLACING BITUMINOUS PAVEMENT



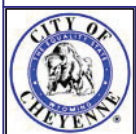
REVISED
2014

STANDARD
DRAWING NO.
02515-01



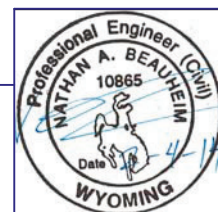
NOTES:

1. MATERIAL COMPACTION SHALL CONFORM TO SECTIONS 02220, 02225, 02226, 02231, AND 02512.
2. WHEN ANY PORTION OF THE PIPE HAS LESS THAN 18" OF COVER TO TOP OF FINISH GRADE OR THE TRENCH IS LESS THAN 30" IN WIDTH THE TRENCH SHALL BE BACKFILLED WITH CITY MIX 3 SLURRY.
3. CONCRETE BASE SHALL BE REINSTALLED WITH #5 EPOXY COATED REBAR, 18 INCHES ON CENTER AND 18" LONG CENTERED IN CONCRETE. MIRITAK FABRIC OR APPROVED EQUAL SHALL BE INSTALLED WITH THE CENTER OVER THE JOINT AND EXTENDED 3" BEYOND SAWCUT END.
4. DEPTH OF BITUMINOUS PATCH SHALL BE A MINIMUM OF 4" OR MATCH EXISTING UP TO 6" ON ALL STREETS DESIGNATED AS LOCAL OR ALLEYS. PATCH DEPTH OF 6" OR MATCH EXISTING UP TO 12" ON ALL STREETS DESIGNATED AS COLLECTOR OR ARTERIALS.
5. EXISTING PAVEMENT SHALL BE CUT BACK A MINIMUM OF 12" BEYOND ANY DISTURBED BASE COURSE, SUBGRADE OR BROKEN ASPHALT. IN ADDITION, THE ADJACENT PAVEMENT SHALL BE MILLED A DEPTH OF 1½" NO LESS THAN ONE FOOT WIDE AND THE TOP LIFT OF PATCH INLAID. SEE SECTION 02515 FOR ADDITIONAL ASPHALT REMOVAL AND PATCHING REQUIREMENTS.



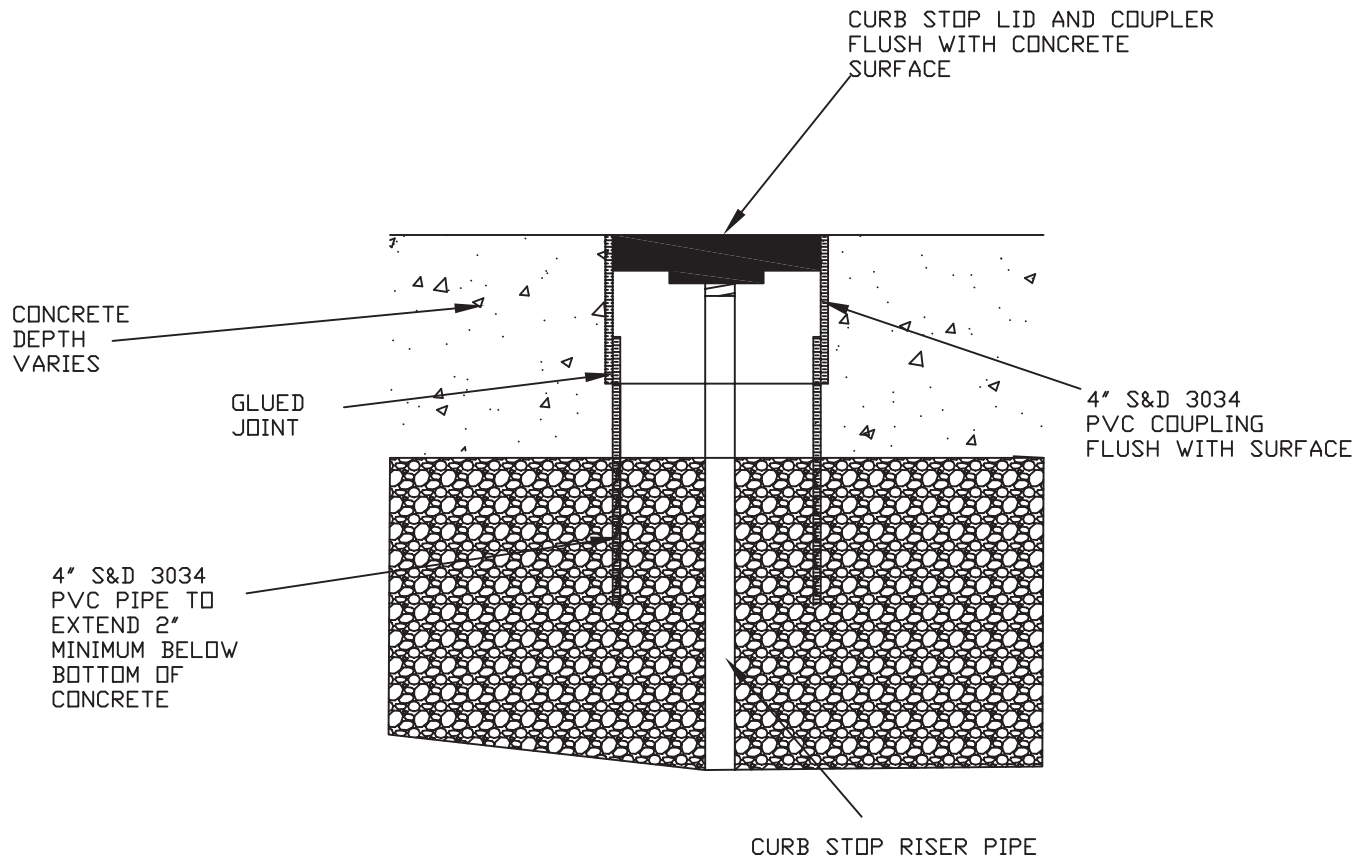
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STANDARD DRAWING

STREET CUT REPAIRS
REPLACING ASPHALT WHERE
CONCRETE BASE EXIST



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2014

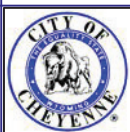
STANDARD
DRAWING NO.
02515-02



TYPICAL CURB STOP ADJUSTMENT

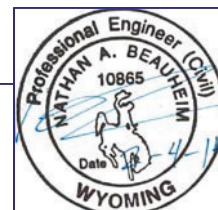
NOTES:

1. ALL CURB STOPS PLACED IN CONCRETE SHALL BE SLEEVED IN ACCORDANCE WITH THIS DRAWING.



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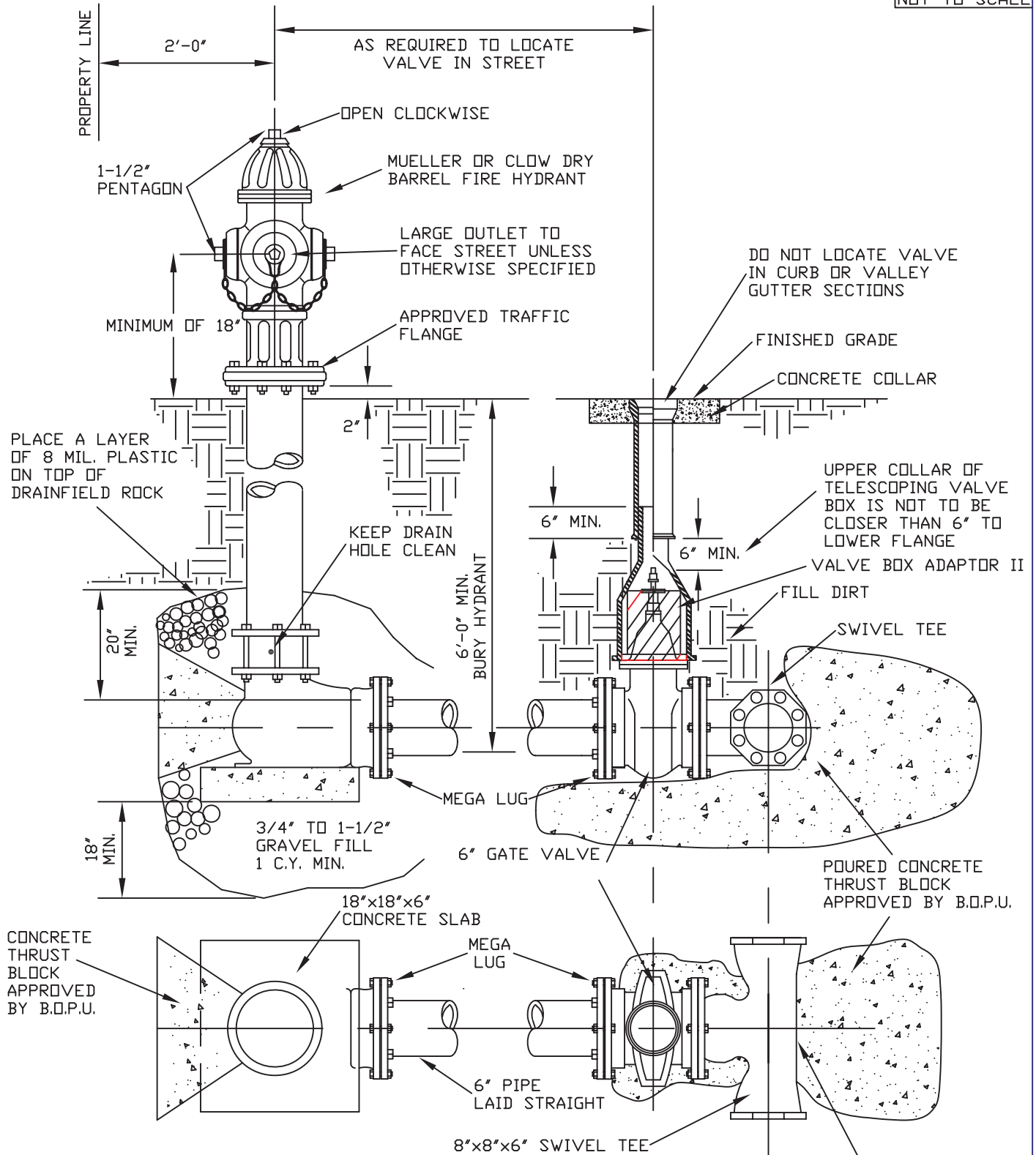
ADJUSTING
CURB STOP IN CONCRETE
DETAIL



NEW
2014

STANDARD
DRAWING NO.
02570-03

NOT TO SCALE



NOTE:

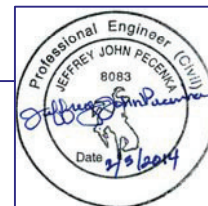
1. DRAIN HOLES ARE TO BE PLUGGED IF LOCATED BELOW WATER TABLE.
2. DOUBLE POLYWRAP ALL UNDERGROUND FACILITIES, INCLUDING THE VALVE BOX, SEE DETAIL 2665-02
3. ALL JOINTS BETWEEN HYDRANT & VALVE MUST BE RESTRAINED.
4. ALL THREAD RESTRAINT IS NOT ALLOWED.

USE BOND BREAKER BETWEEN CONCRETE AND FITTING TYPICAL



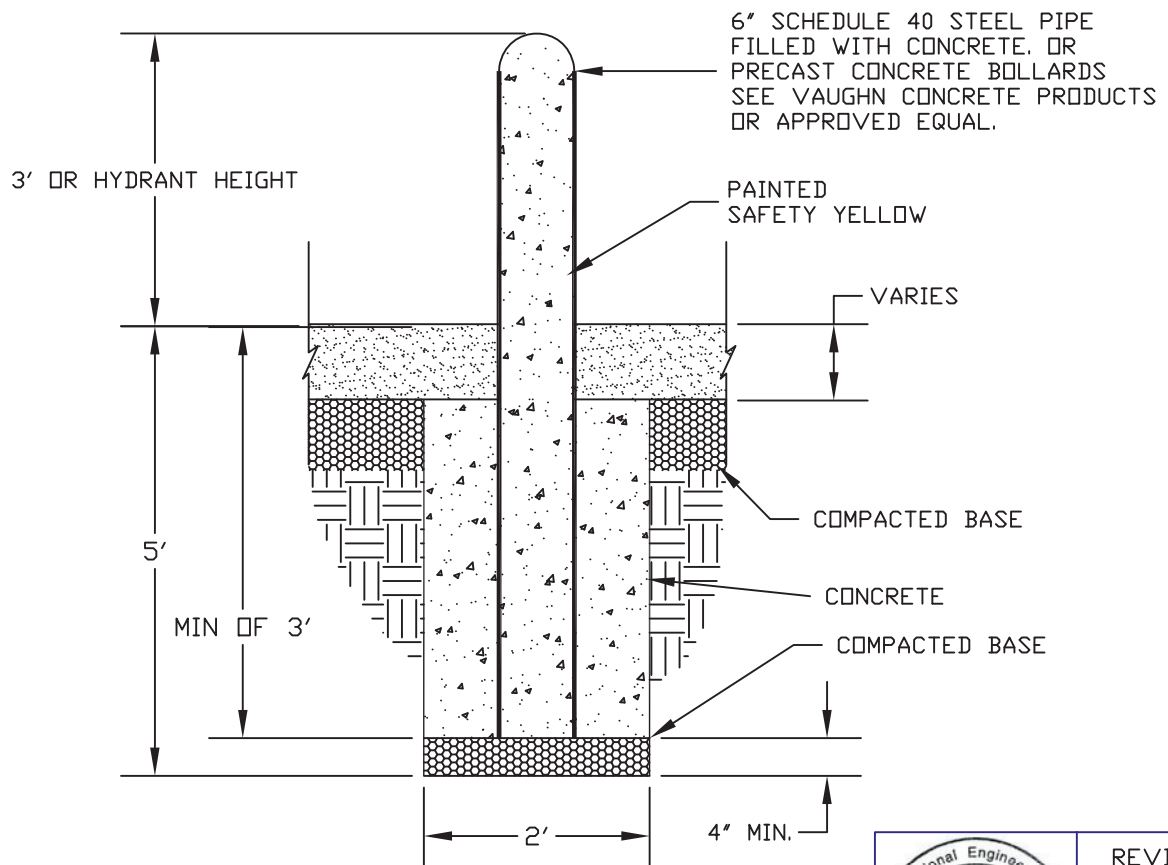
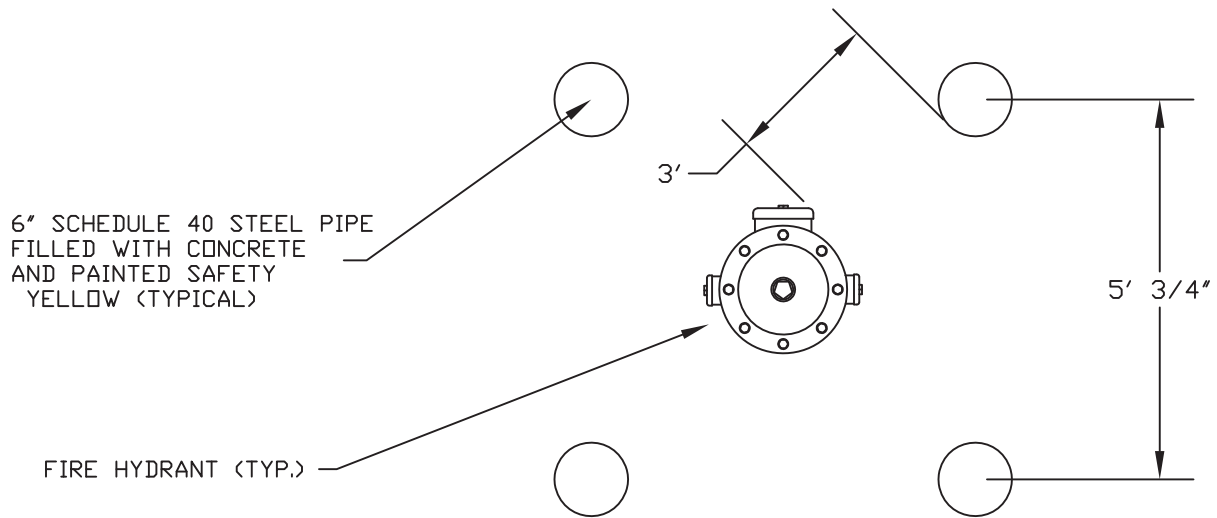
CITY OF CHEYENNE
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TYPICAL FIRE HYDRANT
ASSEMBLY INSTALLATION DETAIL



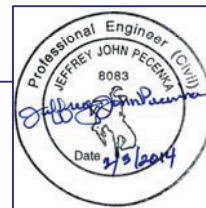
REVISED
2014

STANDARD
DRAWING NO.
02645-01



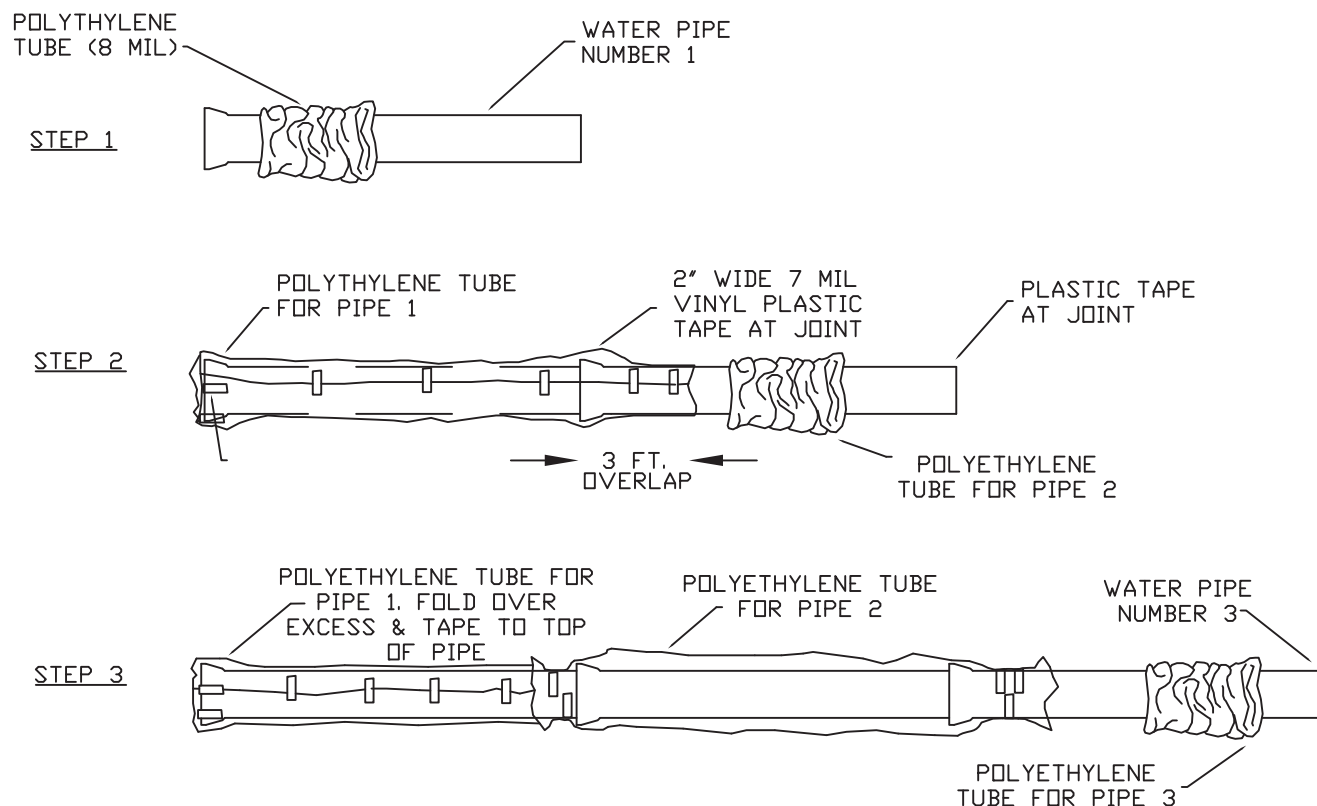
CITY OF CHEYENNE
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UTILITIES
STANDARD DRAWING

BOLLARD DETAIL



REVISED
2014

STANDARD
DRAWING NO.
02645-02

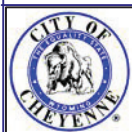


FIELD INSTALLATION - POLYETHYLENE WRAP

- STEP 1** PLACE TUBE OF 8 MIL THICKNESS, ANSI AWWA C105/A215 V-BID FILM POLYETHYLENE TUBE MATERIAL ON PIPE JUST PRIOR TO LOWERING IT INTO THE TRENCH. NOT WHEN IT IS STOCKPILED.
- STEP 2** PULL THE TUBE OVER THE LENGTH OF THE PIPE. TAPE TUBE TO JOINT. FOLD MATERIAL AROUND THE ADJACENT SPIGOT END AND WRAP WITH TAPE TO HOLD PLASTIC IN PLACE.
- STEP 3** OVERLAP FIRST TUBE WITH ADJACENT TUBE OF 4 MIL CROSS LAMINATED POLYETHYLENE TUBE ENCASUREMENT AND SECURE WITH PLASTIC ADHESIVE TAPE, THE POLYETHYLENE TUBE MATERIAL COVERING THE PIPE SHALL BE LOOSE, EXCESS MATERIAL SHALL BE NEATLY DRAWN AROUND THE PIPE BARREL, FOLDED ON TOP OF THE PIPE AND TAPED IN PLACE WITH 3 FOOT OVERLAPS.

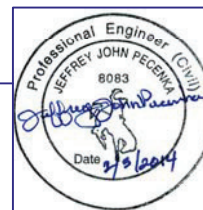
NOTE:

DUCTILE IRON PIPE AND RELATED CONSTRUCTION MATERIALS SHALL BE DOUBLE WRAPPED.



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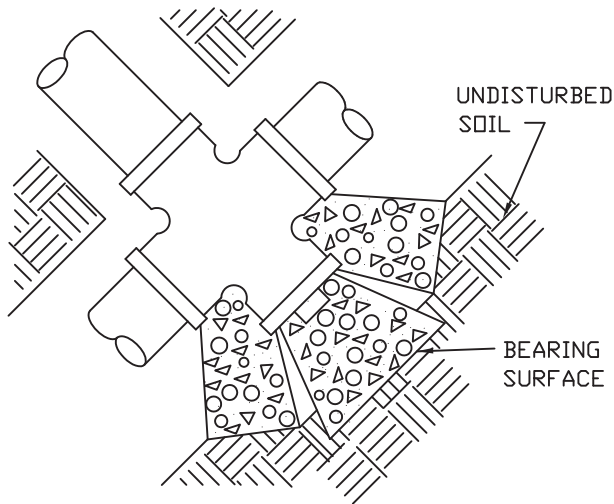
POLYETHYLENE WRAP



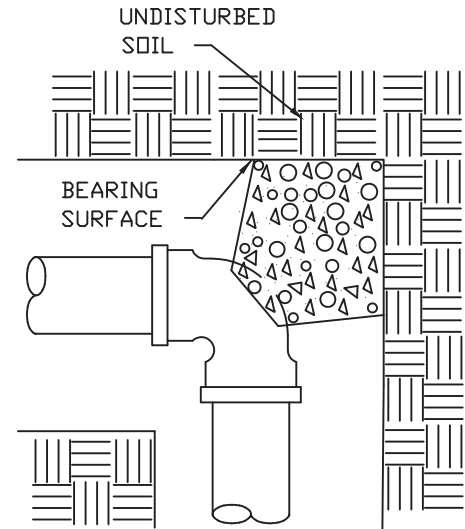
REVISED
2014

STANDARD
DRAWING NO.
02665-02

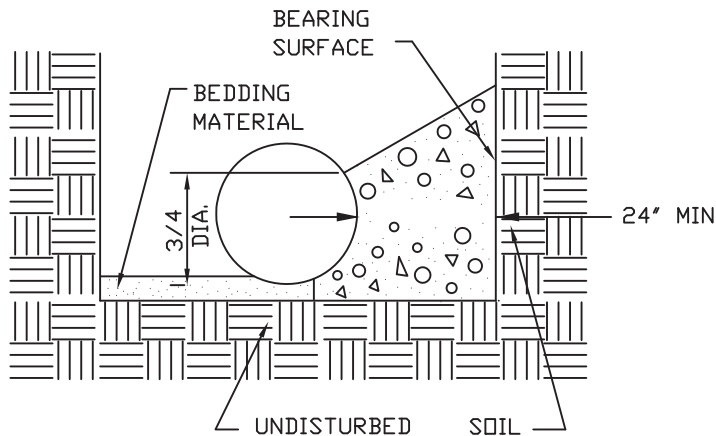
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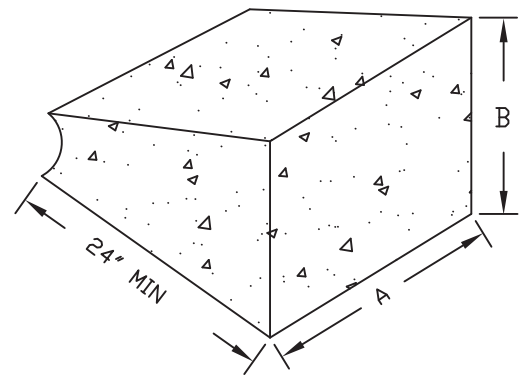
SINGLE PLUGGED CROSS



HORIZONTAL BEND



TYPICAL CROSS SECTION



THRUST BLOCK

MINIMUM DIMENSIONS FOR THRUST BLOCKING

FITTING SIZE	TEES & PLUGS		90° BENDS		45° BENDS & WYES		REDUCERS & 22 1/2° BENDS		11 1/4° BENDS	
	A	B	A	B	A	B	A	B	A	B
4"	1'-8"	1'-6"	2'-0"	1'-6"	2'-0"	1'-6"	1'-6"	0'-9"	1'-6"	0'-9"
6"	2'-8"	2'-0"	3'-3"	2'-9"	2'-6"	2'-0"	2'-6"	1'-0"	1'-6"	0'-6"
8"	3'-6"	3'-0"	4'-3"	3'-6"	3'-8"	2'-6"	2'-3"	2'-0"	1'-6"	1'-0"
10"	4'-3"	4'-0"	4'-6"	5'-3"	4'-0"	3'-3"	3'-0"	2'-3"	1'-6"	1'-0"
12"	5'-6"	4'-6"	6'-8"	5'-0"	4'-6"	4'-0"	3'-6"	2'-9"	2'-0"	1'-0"

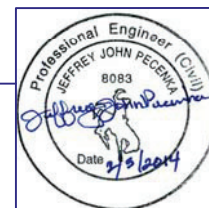
NOTE:

1. FITTINGS TO BE SEPARATED FROM BLOCKS WITH AN APPROVED BOND BREAKER, SUCH AS POLY WRAP.
2. ALL BLOCKS TO BEAR AGAINST UNDISTURBED MATERIAL.
3. ALL CONCRETE SHALL HAVE A MINIMUM STRENGTH OF 4,000 PSI.
4. FOR FIRE HYDRANTS, THRUST BLOCKS A = 39", B = 33", LENGTH = 36"



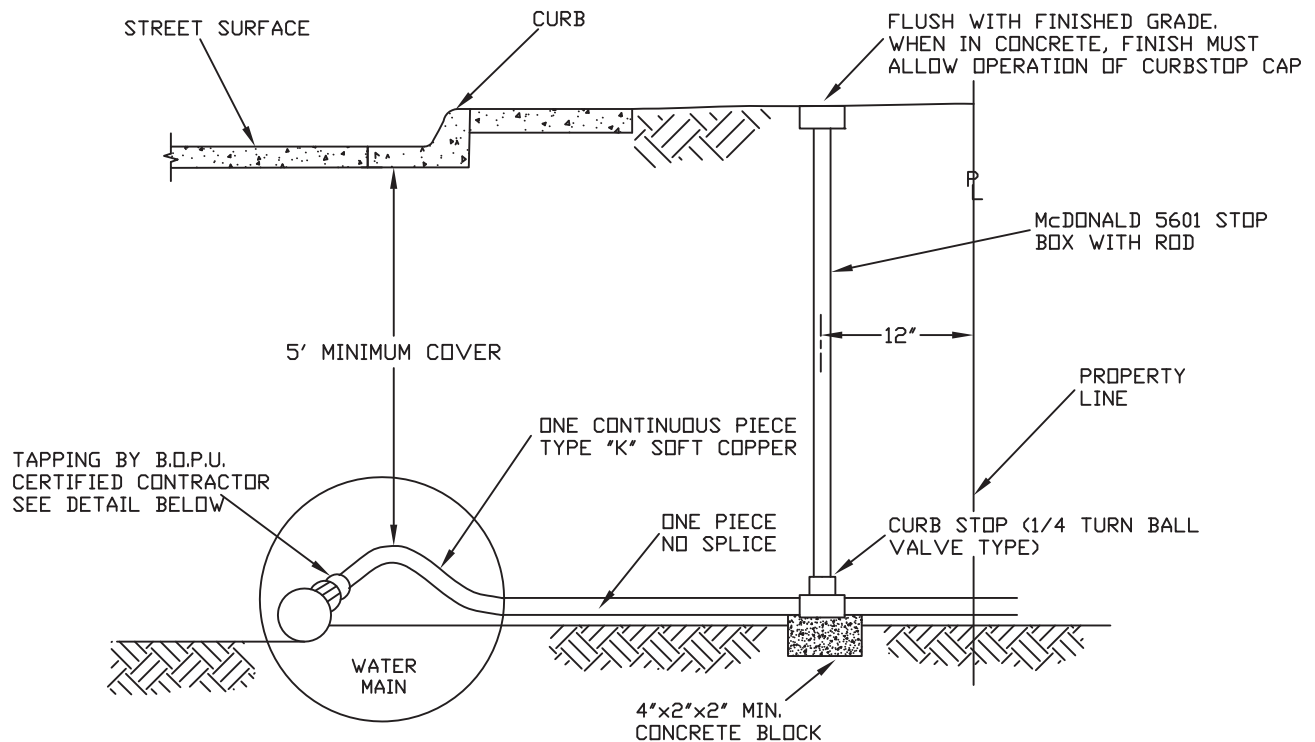
CITY OF CHEYENNE
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STANDARD DRAWING

MINIMUM THRUST BLOCK
REQUIREMENTS



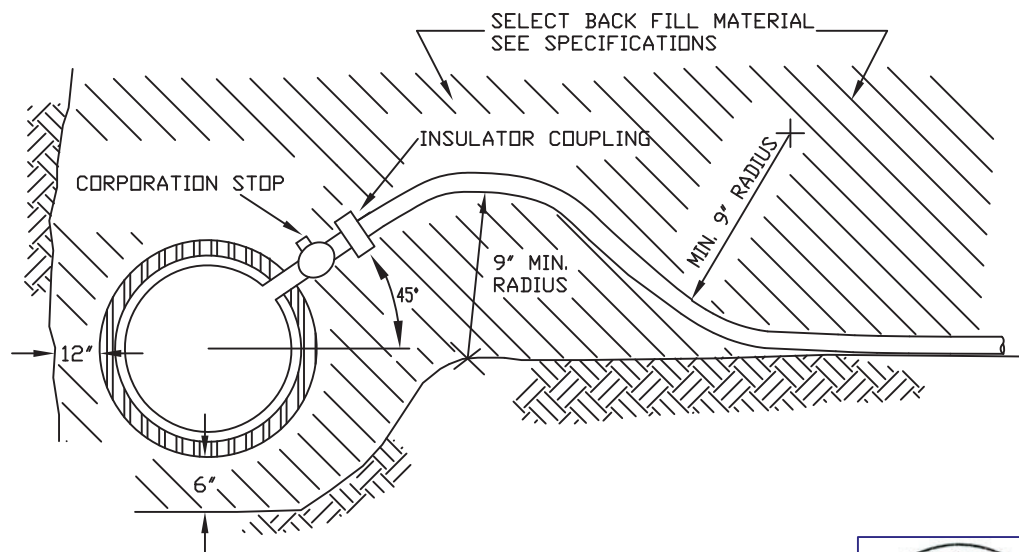
REVISED
2014

STANDARD
DRAWING NO.
02665-04



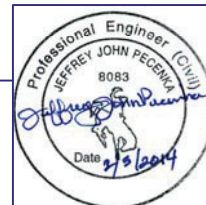
NOTE:

1. B.O.P.U. MAKES TAPS 3" AND LARGER.
2. THIS DRAWING IS FOR TAPPING DUCTILE IRON PIPE. SEE PVC SPECIFICATIONS FOR TAPPING PVC PIPE.
3. ALL METERS PITS SHALL BE LOCATED A MIN. OF 2' (FEET) AND A MAX. OF 5' (FEET) INSIDE PROPERTY LINE OR AS APPROVED ON PLANS.
4. 1½" TO 2" WILL BE ALLOWED TO TAP AT 90 DEGREES TO MAIN.
5. ALL METERS 1½" AND LARGER SHALL BE INSTALLED IN PITS.



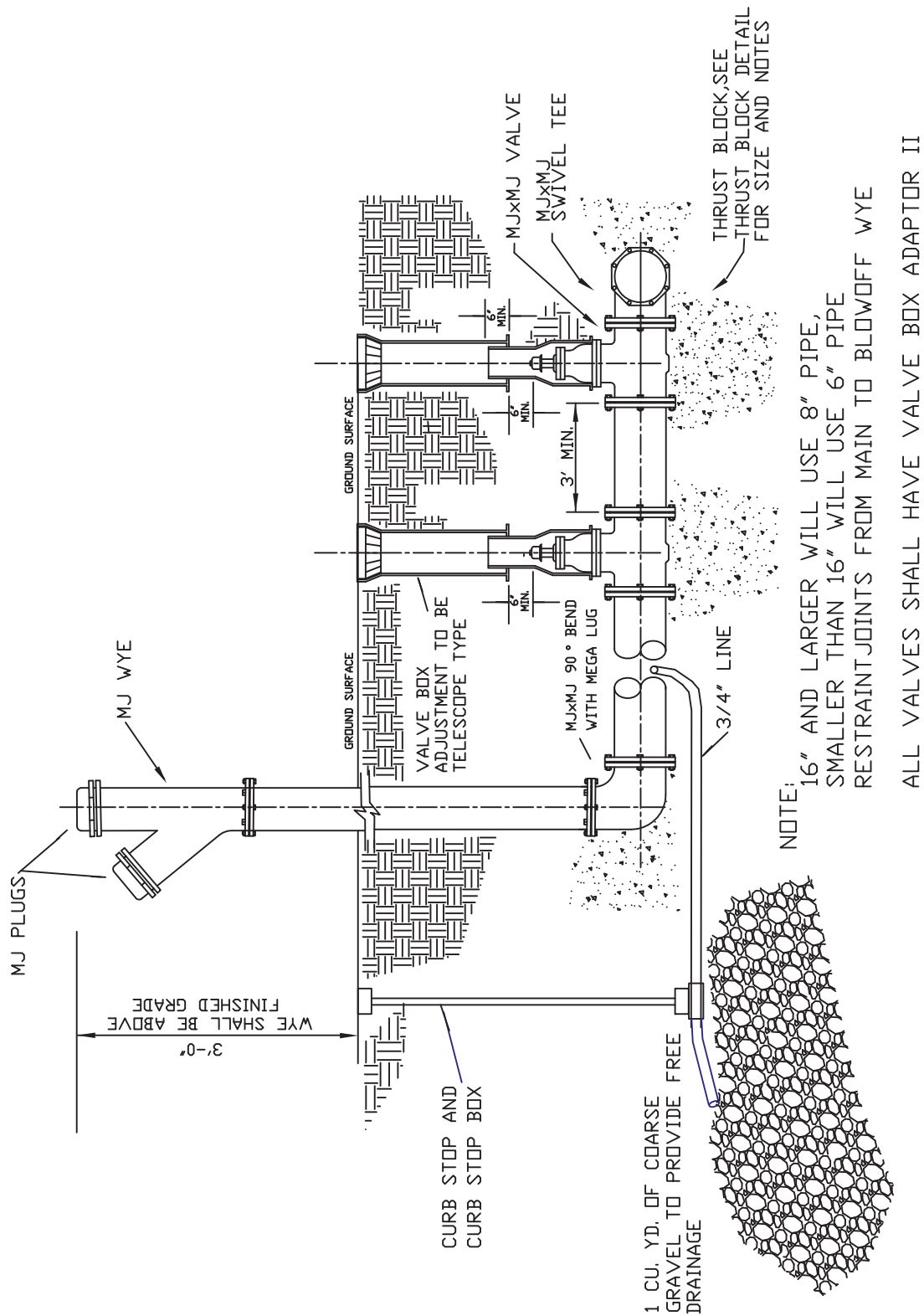
CITY OF CHEYENNE
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WATER SERVICE LINE AND TAP



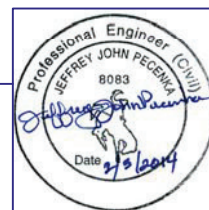
REVISED
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STANDARD
DRAWING NO.
02665-06



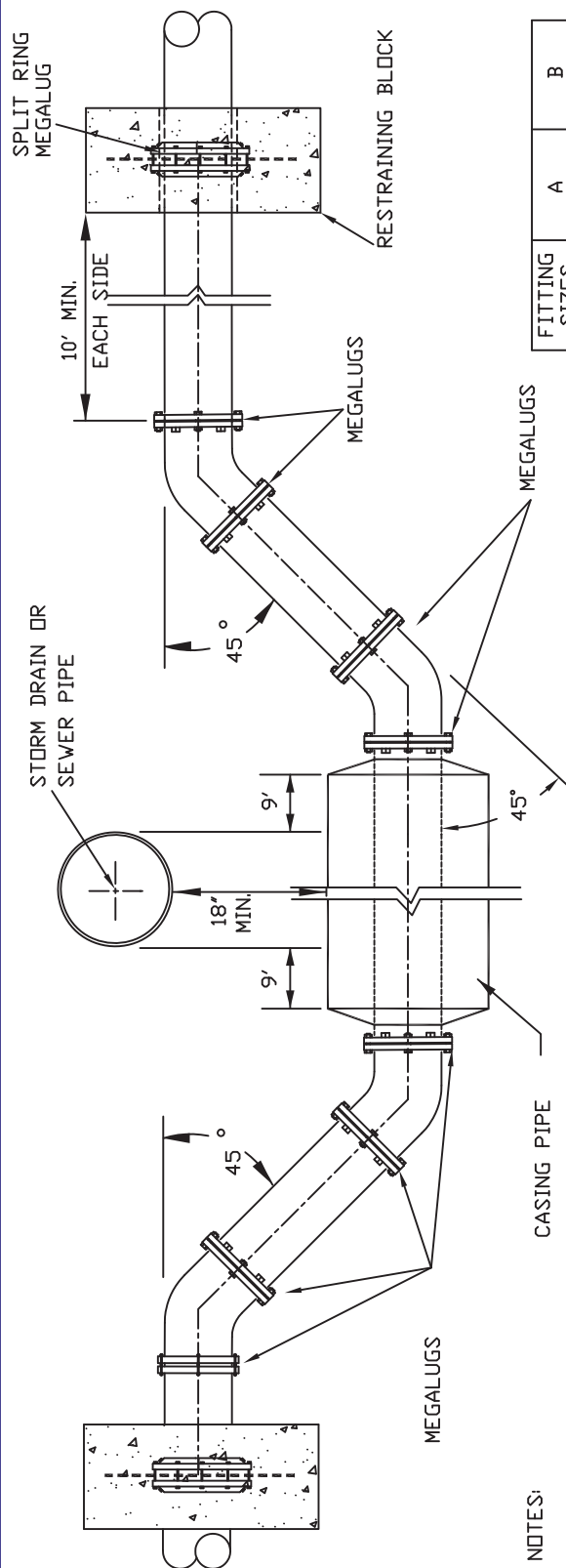
CITY OF CHEYENNE
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STANDARD DRAWING

6" BLOW OFF DETAIL



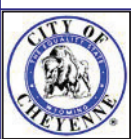
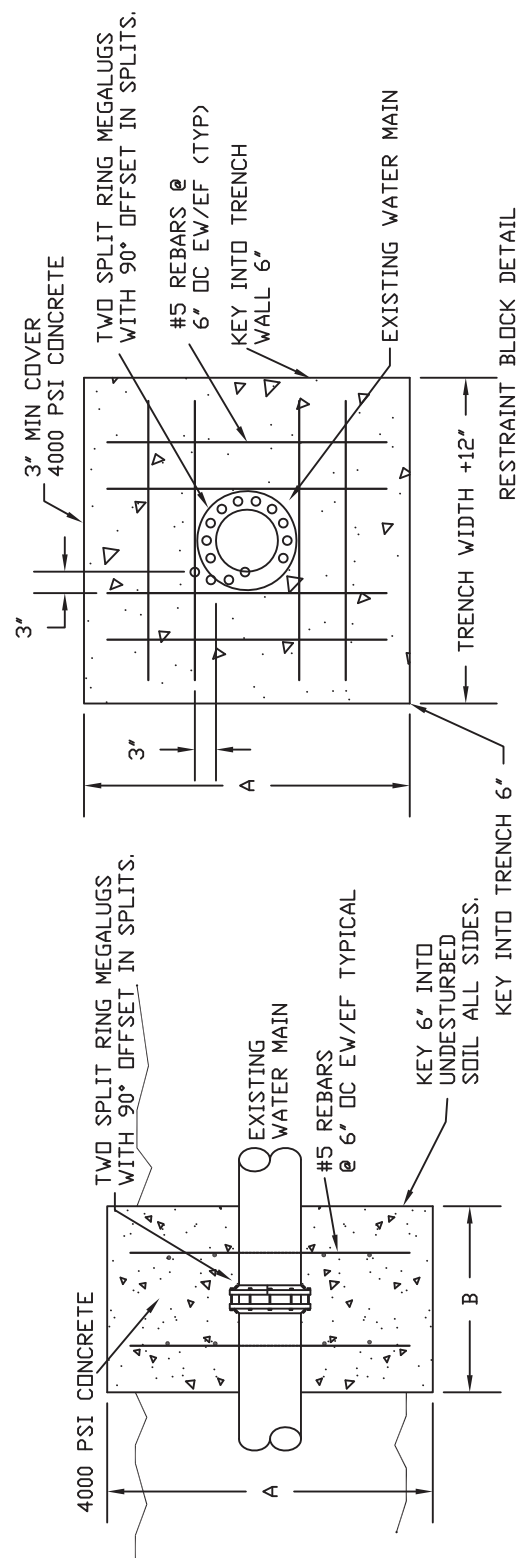
REVISED
2014

STANDARD
DRAWING NO.
02665-08



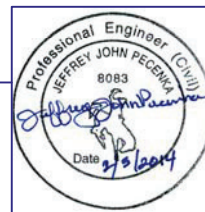
FITTING SIZES	A	B
4"	2'-4"	2'-0"
6"	2'-4"	2'-6"
8"	3'-6"	3'-3"
10"	4'-3"	4'-0"
12"	5'-6"	4'-6"

- NOTES:
1. THRUST BLOCKS MAY BE REQUIRED IN UNSTABLE MATERIAL.
 2. PER D.E.Q. REQUIREMENTS, A CASING PIPE OR APPROPRIATE PIPE ENCASEMENT, SHALL BE REQUIRED WHEN WATER MAIN IS LESS THAN 18" ABOVE A SANITARY OR STORM SEWER.
 3. JOINTS INSIDE THE CASING PIPE OR THE PIPE ENCASEMENT MUST ALSO BE RESTAINED JOINTS.
 4. SUPPORT SANITARY SEWER OVER WATER MAIN.
 5. SEE DRAWING 02665-01 FOR DETAILS ON CASING PIPE SPACERS AND END SEALS.

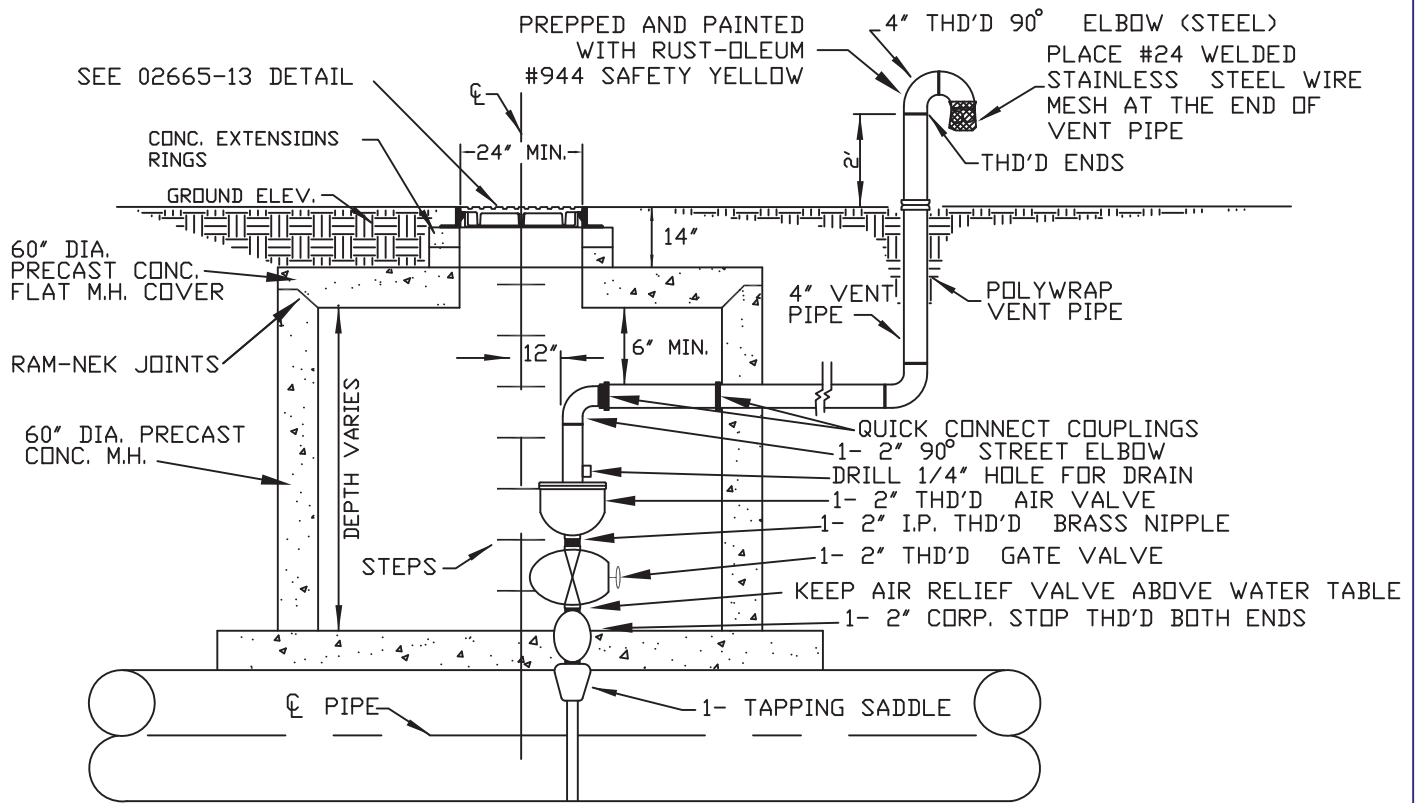


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WATER MAIN ADJUSTMENT



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02665-09

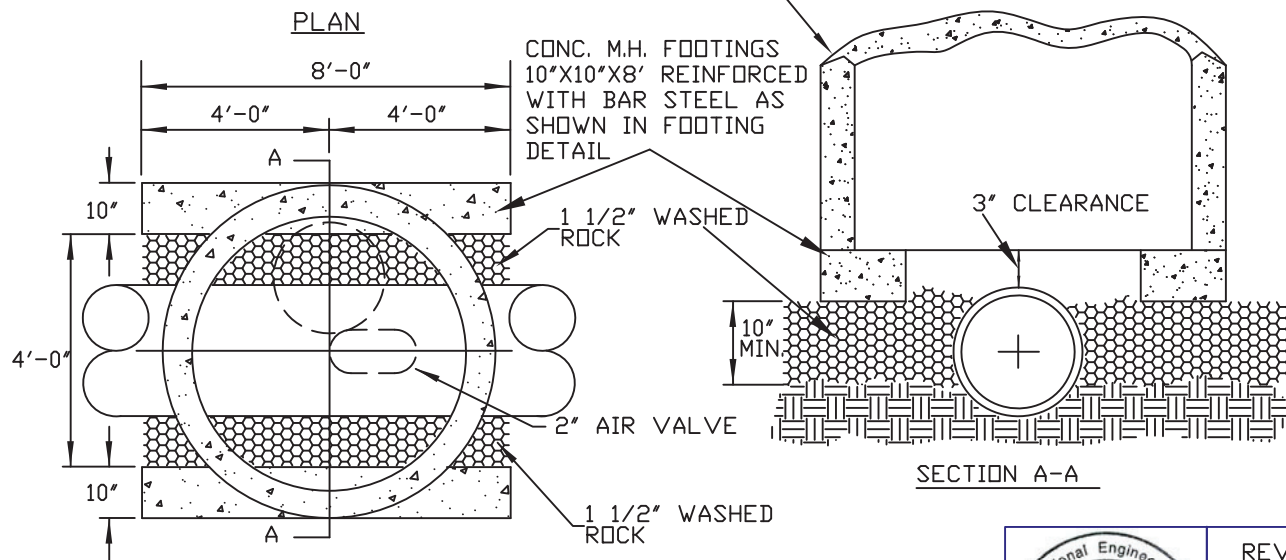
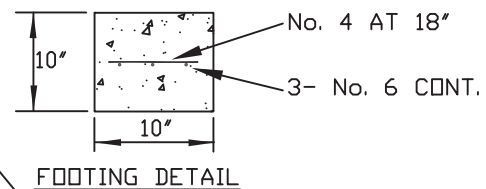


ELEVATION

NOTE:

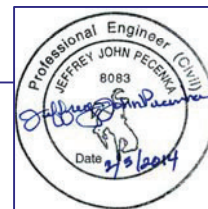
1. AIR VALVE SHALL BE VAL-MATIC OR APCO.

* PRECAST MANHOLE SECTIONS SHALL CONFORM TO ASTM C 478



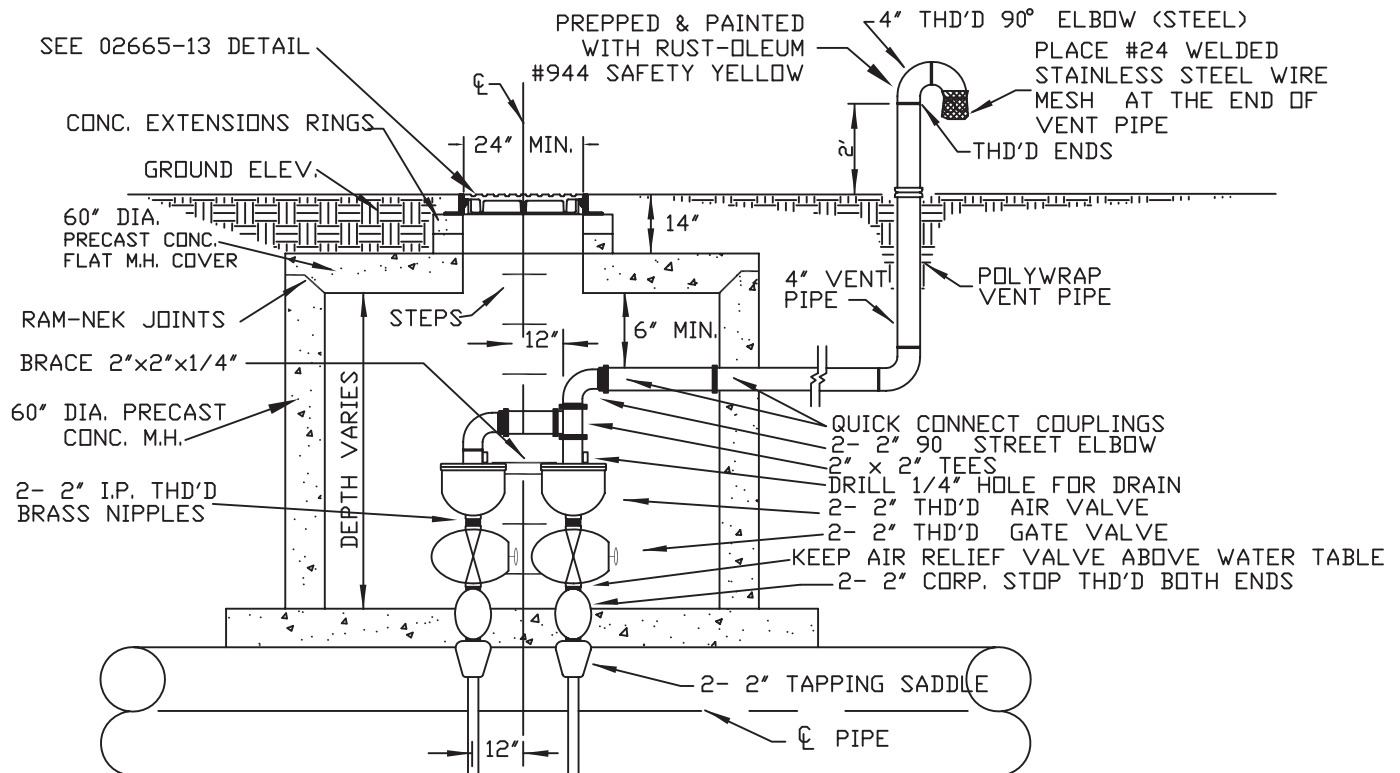
CITY OF CHEYENNE
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STANDARD AIR VALVE FOR MAINS 10 INCH AND SMALLER



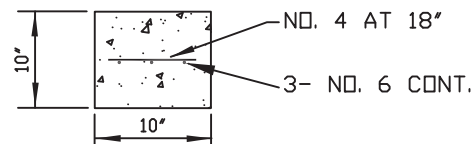
REVISED
2014

STANDARD
DRAWING NO.
02665-10

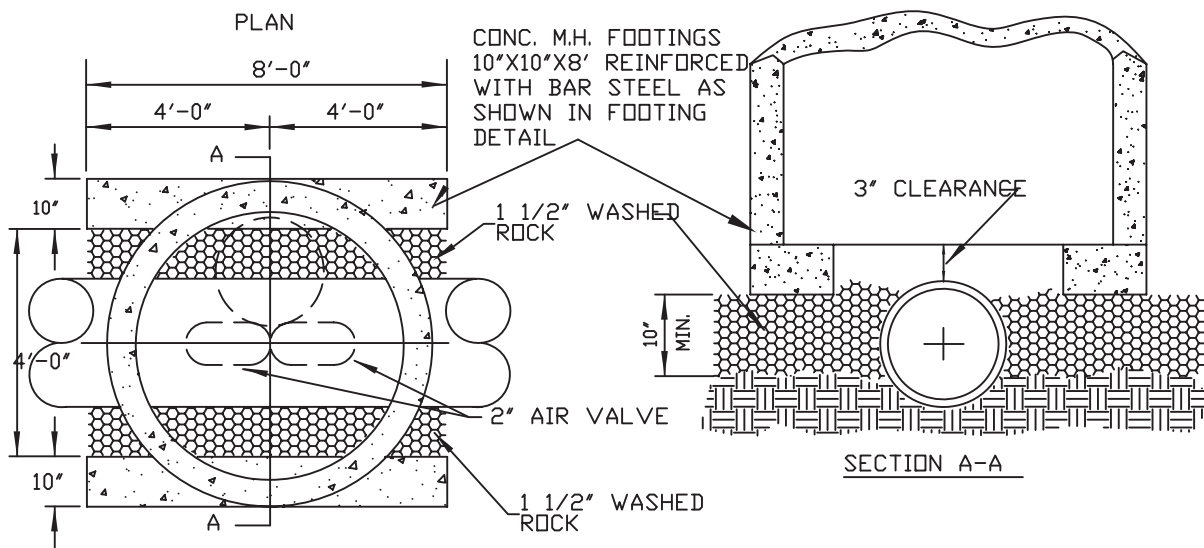


ELEVATION

NOTE:
1. AIR VALVE SHALL BE VAL-MATIC OR APCO.

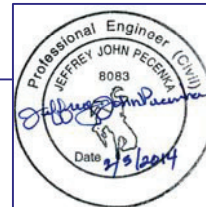


FOOTING DETAIL



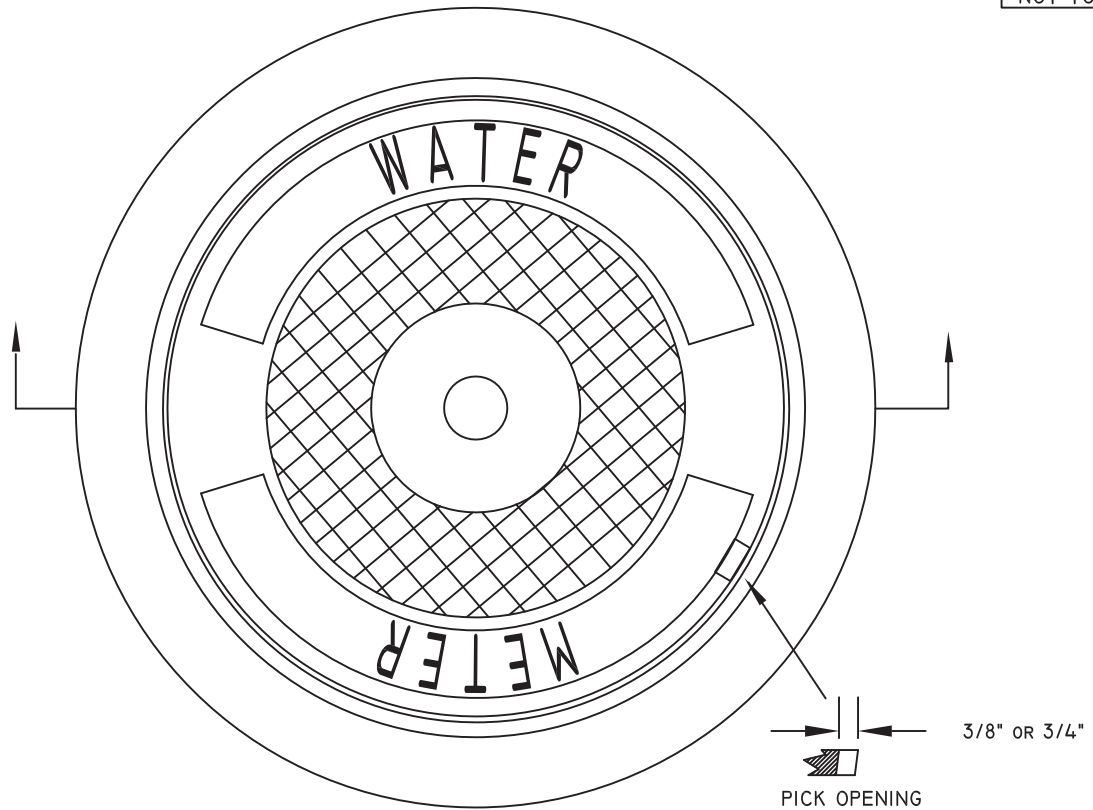
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STANDARD DRAWING

STANDARD AIR VALVE FOR
MAINS 12 INCH AND LARGER

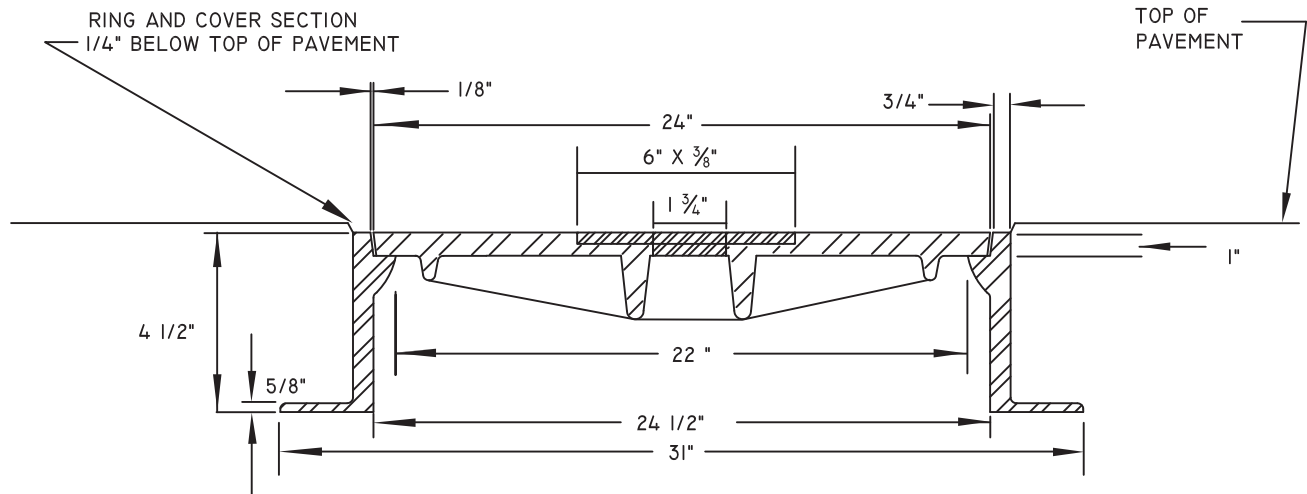


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2014

STANDARD
DRAWING NO.
02665-11



RING AND COVER SECTION



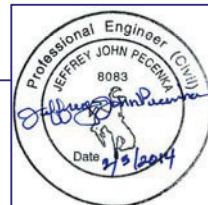
NOTES:

1. WITH APPROVED BOLT DOWN COVER AND WATER TIGHT GASKET IN FLOOD PLAIN AND OTHER AREAS SUBJECT TO INUNDATION.
2. CASTING INC. CI.-24-R / AL- 24-R OR APPROVED EQUAL. MUST MEET HS-20-44 LOADING.
3. WEIGHT OF CASTING LIDS SHALL BE A MINIMUM OF 135 POUND COVER
4. $1\frac{3}{4}$ \" HOLE IN CENTER OF LID WITH A $6\" \times \frac{3}{8}\"$ RECESS IN CENTER OF LID FOR A NEPTUNE PITANTAN 12527-000 RADIO READ OUT



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BOARD OF PUBLIC UTILITIES
STANDARD DRAWING

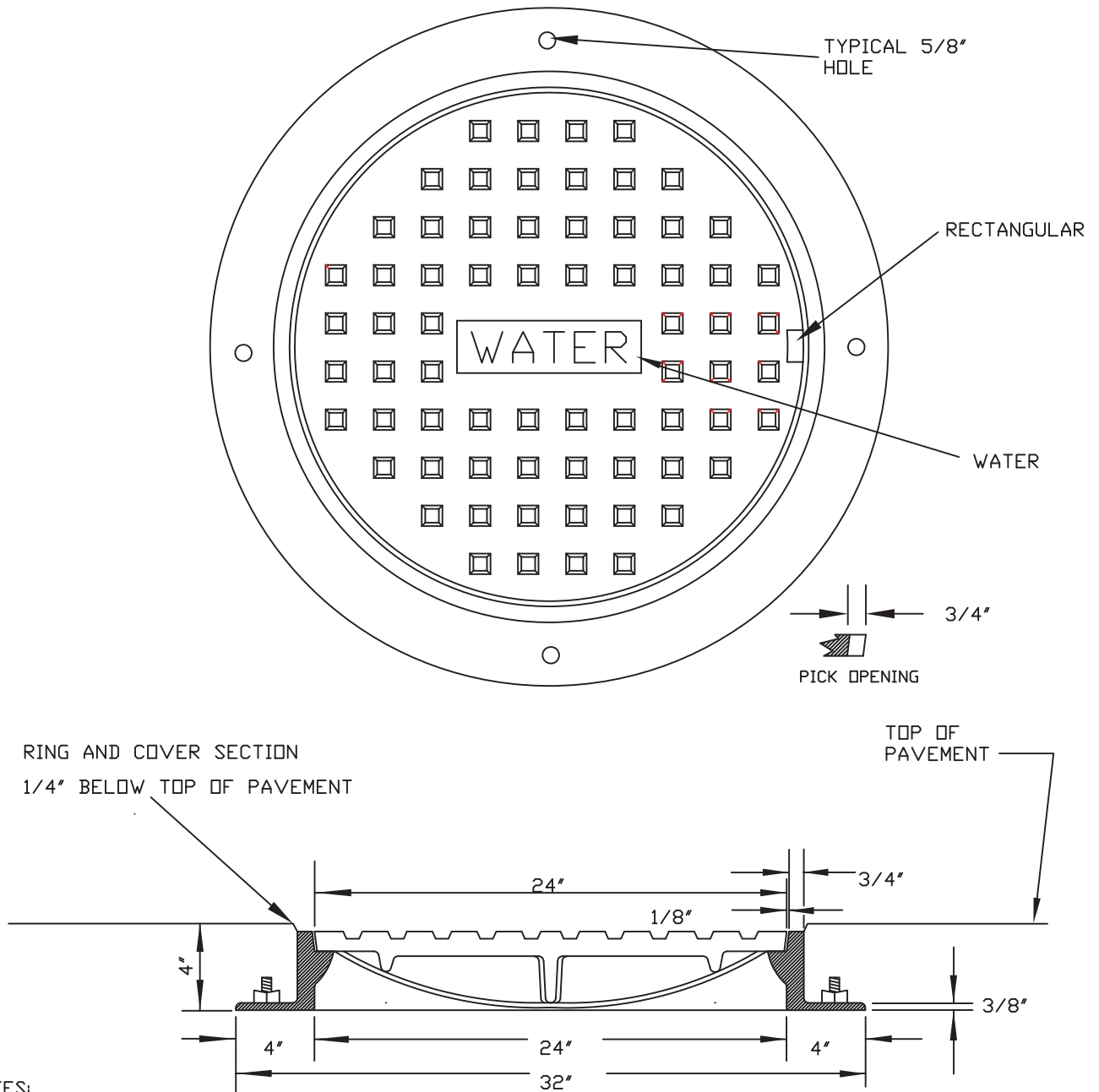
TRAFFIC AREA METER PIT
RING AND COVER DETAILS



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2014

STANDARD
DRAWING NO.
02665-12

NOT TO SCALE



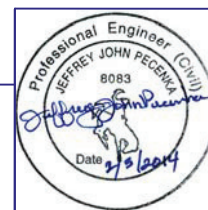
NOTES:

1. FIT BETWEEN RING AND COVER MUST BE MACHINED.
2. MUST HAVE APPROVED BOLT DOWN COVER AND GASKET IN FLOOD PLAIN AND OTHER AREAS SUBJECT TO INUNDATION.
3. MANHOLES OUTSIDE PAVED STREETS OR ALLEYS SHALL HAVE BOLT TYPE COVER THAT BOLTS TO THE RING AND THE RING SHALL BE BOLTED TO CONE SECTION OR FLAT TOP SECTION OF MANHOLE. USE FOUR (4) $\frac{1}{2}$ " DIAMETER X 3" LONG STAINLESS STEEL RAM SET ANCHORS OR APPROVED EQUAL.
4. EAST JORDAN IRON WORKS 330-4 RING WITH A CHECKERED FLUSH MOUNT COVER MARKED "WATER" OR APPROVED EQUAL.
5. USE A MINIMUM 135 POUND COVER.



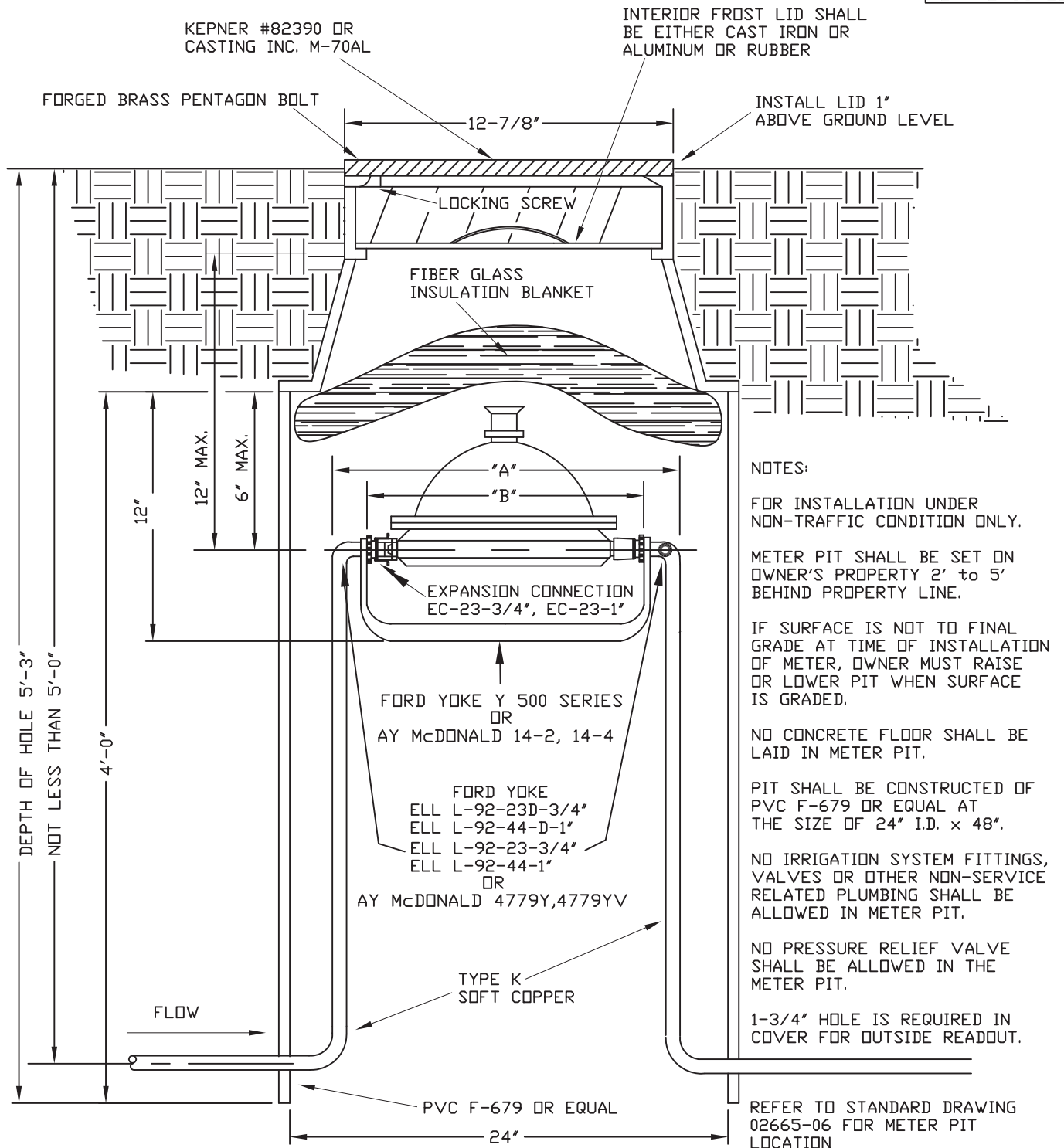
CITY OF CHEYENNE
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BOARD OF PUBLIC
UTILITIES
STANDARD DRAWING

AIR VAULT RING AND COVER
DETAILS



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2014
STANDARD
DRAWING NO.
02665-13

NOT TO SCALE



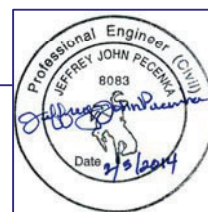
METER SIZE	"A"	"B"
3/4"	14-1/4"	10-5/8"
1"	17-1/4"	12-5/8"

WARNING! INSTALLATION OF A BACK FLOW PREVENTER REQUIRES INSTALLATION OF AN EXPANSION TANK IN ACCORDANCE WITH THE MANUFACTURER'S SPECIFICATIONS.



CITY OF CHEYENNE
AND
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STANDARD DRAWING

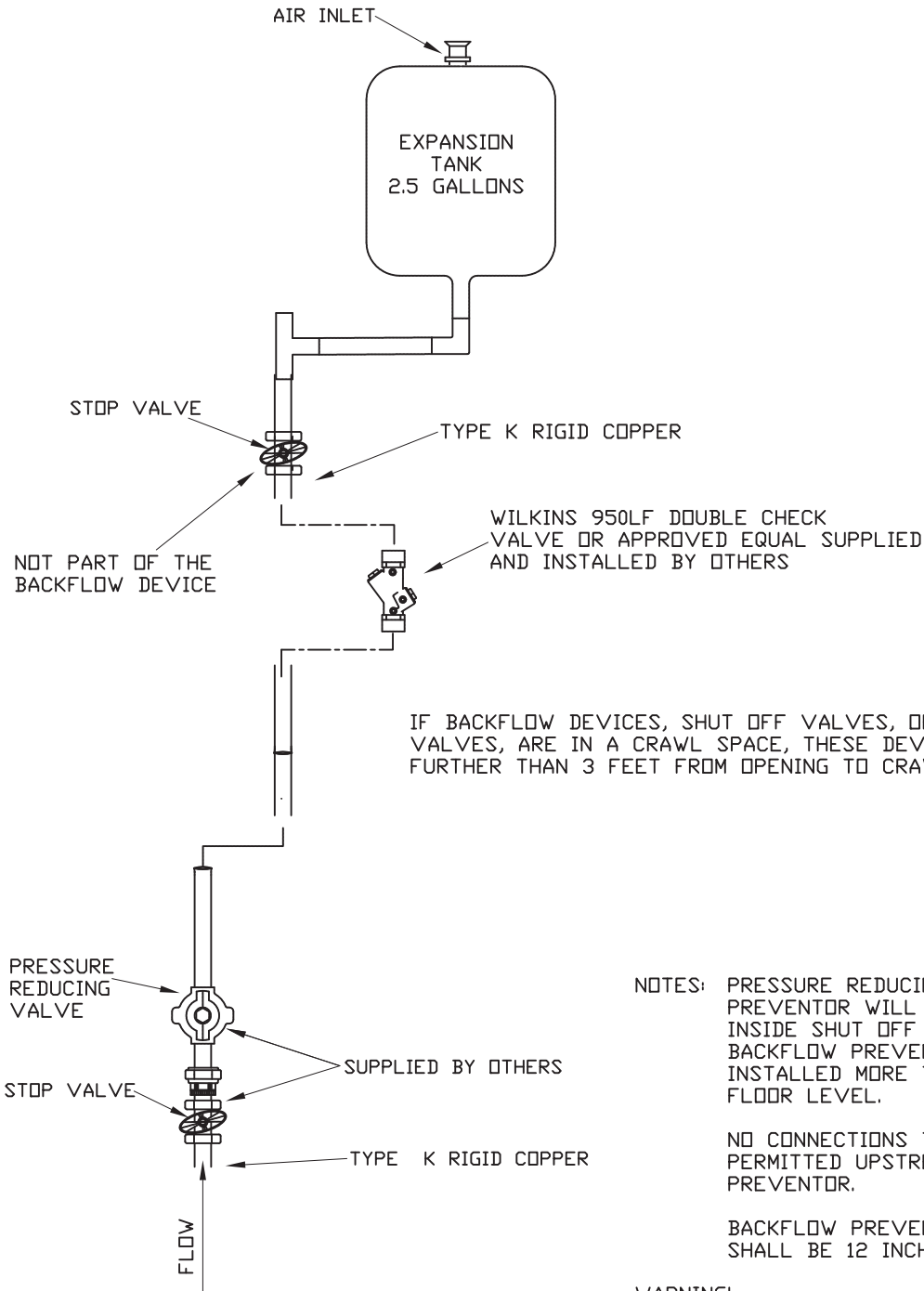
NON-TRAFFIC INSTALLATION
OUTSIDE METER PIT DETAIL
FOR 3/4" AND 1" METERS



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STANDARD
DRAWING NO.
02665-14

NOT TO SCALE



IF BACKFLOW DEVICES, SHUT OFF VALVES, OR PRESSURE REDUCING VALVES, ARE IN A CRAWL SPACE, THESE DEVICES SHALL BE NO FURTHER THAN 3 FEET FROM OPENING TO CRAWL SPACE

NOTES: PRESSURE REDUCING VALVE AND BACKFLOW PREVENTOR WILL BE INSTALLED AFTER INSIDE SHUT OFF VALVE. BACKFLOW PREVENTOR SHALL NOT BE INSTALLED MORE THAN FOUR FEET OFF FLOOR LEVEL.

NO CONNECTIONS TO THE SERVICE SHALL BE PERMITTED UPSTREAM OF THE BACKFLOW PREVENTOR.

BACKFLOW PREVENTOR SET CLEARANCE SHALL BE 12 INCHES ON ALL SIDES.

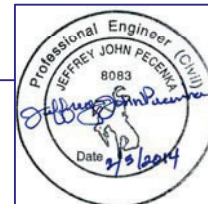
WARNING! INSTALLATION OF A BACK FLOW PREVENTOR REQUIRES INSTALLATION OF AN EXPANSION TANK IN ACCORDANCE WITH THE MANUFACTURER'S SPECIFICATIONS.

METER SIZE	WILKINS DOUBLE CHECK VALVE
3/4" & 1"	950LF OR EQUAL



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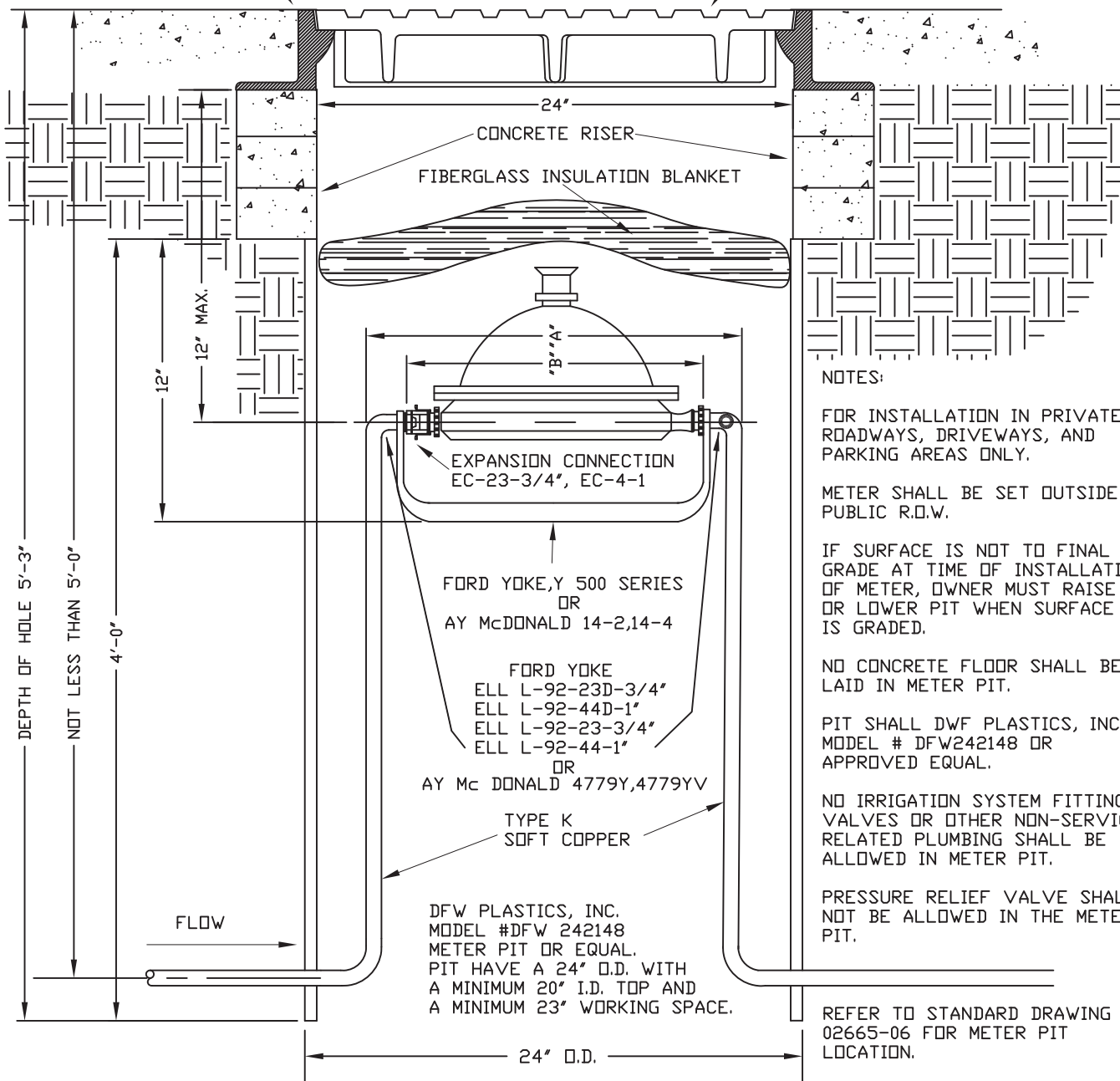
INSIDE INSTALLATION OF
BACKFLOW PREVENTER W/
OUTSIDE METER PIT FOR
3/4" AND 1" METER DETAILS



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STANDARD
DRAWING NO.
02665-15

INSTALL LID TO MATCH ASPHALT
OR CONCRETE ELEVATION

SEE NEW DETAIL 02665-12
TRAFFIC AREA METER PIT
RING AND COVER



NOTES:

FOR INSTALLATION IN PRIVATE
ROADWAYS, DRIVEWAYS, AND
PARKING AREAS ONLY.

METER SHALL BE SET OUTSIDE
PUBLIC R.O.W.

IF SURFACE IS NOT TO FINAL
GRADE AT TIME OF INSTALLATION
OF METER, OWNER MUST RAISE
OR LOWER PIT WHEN SURFACE
IS GRADED.

NO CONCRETE FLOOR SHALL BE
LAID IN METER PIT.

PIT SHALL DFW PLASTICS, INC.
MODEL # DFW242148 OR
APPROVED EQUAL.

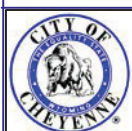
NO IRRIGATION SYSTEM FITTINGS,
VALVES OR OTHER NON-SERVICE
RELATED PLUMBING SHALL BE
ALLOWED IN METER PIT.

PRESSURE RELIEF VALVE SHALL
NOT BE ALLOWED IN THE METER
PIT.

REFER TO STANDARD DRAWING
02665-06 FOR METER PIT
LOCATION.

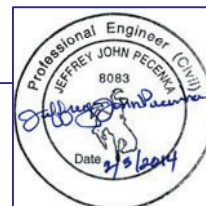
WARNING: INSTALLATION OF A BACK
FLOW PREVENTER REQUIRES
INSTALLATION OF AN EXPANSION
TANK IN ACCORDANCE WITH THE
MANUFACTURER'S SPECIFICATIONS.

METER SIZE	"A"	"B"
3/4"	14 1/4"	10-5/8"
1"	17 1/4"	12-5/8"



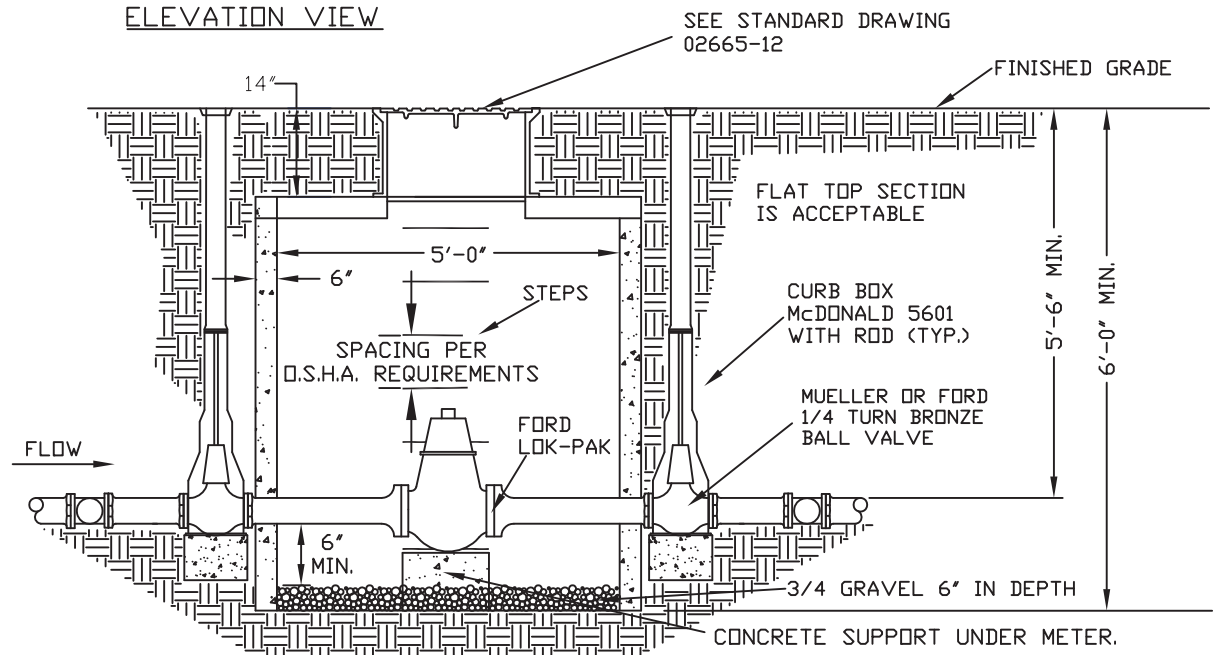
CITY OF CHEYENNE
AND
BOARD OF PUBLIC
UTILITIES
STANDARD DRAWING

TRAFFIC INSTALLATION OUTSIDE
METER PIT DETAIL FOR 3/4" AND 1"
METERS



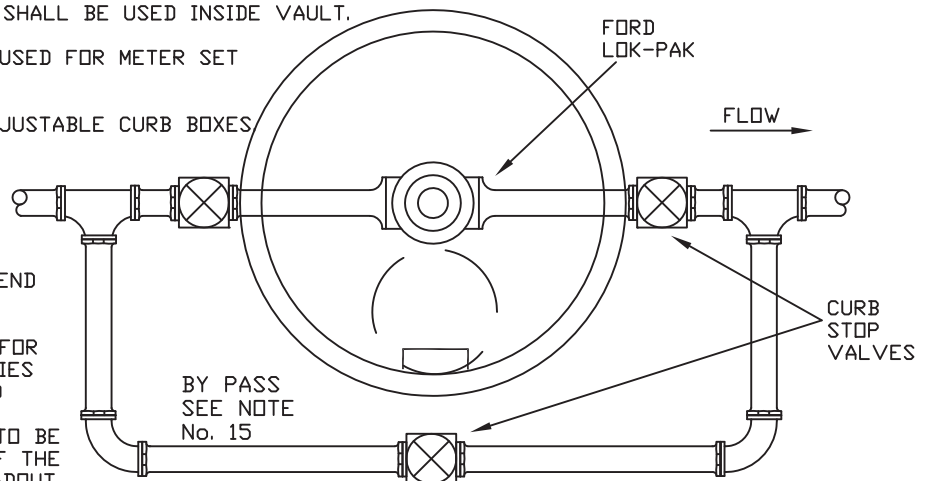
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STANDARD
DRAWING NO.
02665-16



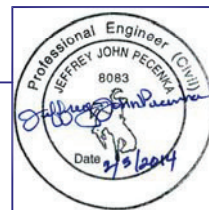
NOTES:

- IF SURFACE IS NOT TO FINAL GRADE AT THE TIME OF INSTALLATION OF METER, OWNER MUST ADJUST PIT TO FINISH GRADE.
- METER PIT TO BE LOCATED ON PRIVATE PROPERTY, CURB STOP TO BE LOCATED IN THE RIGHT OF WAY.
- METER SETTING MUST BE INSPECTED PRIOR TO BACKFILLING.
- NO CONCRETE IS TO BE PLACED IN FLOOR OF METER PIT. 3/4" ROCK SHALL BE INSTALLED ON PIT FLOOR TO A DEPTH OF 6".
- MATERIALS USED FOR SERVICE LINE PIPING THROUGH PIT SHALL BE 1-1/2" OR 2" TYPE "K" SOFT COPPER 20'S. COILED SOFT COPPER IS NOT ALLOWED THROUGH METER PIT. ALL FITTINGS SHALL BE FLARED OR MUELLER 110 COMPRESSION FITTINGS.
- IF GROUND CONDITIONS ARE UNSTABLE, FOOTINGS SHALL BE REQUIRED.
- THE WORD "WATER" SHALL BE CAST ON THE LID.
- COPPER COMPRESSION JOINTS SHALL BE USED INSIDE VAULT.
- FORD CF LOK-PAK SHALL BE USED FOR METER SET INSIDE PIT.
- ALL VALVES SHALL HAVE ADJUSTABLE CURB BOXES
- RIGID COPPER AND SILVER SOLDER IS ALLOWED FOR METER PITS.
- MANHOLE STEPS SHALL BEGIN 2'-4" MAX. FROM COVER AND END 1'-6" MAX. FROM FLOOR.
- A BYPASS IS ONLY ALLOWED FOR MEDICAL AND DENTAL FACILITIES UNLESS OTHERWISE APPROVED
- A 1-3/4" DIAMETER HOLE IS TO BE DRILLED NEAR THE CENTER OF THE MANHOLE COVER FOR THE READOUT.



CITY OF CHEYENNE
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UTILITIES
STANDARD DRAWING

WATER METER PIT DETAIL FOR 1 1/2" and 2" LINES

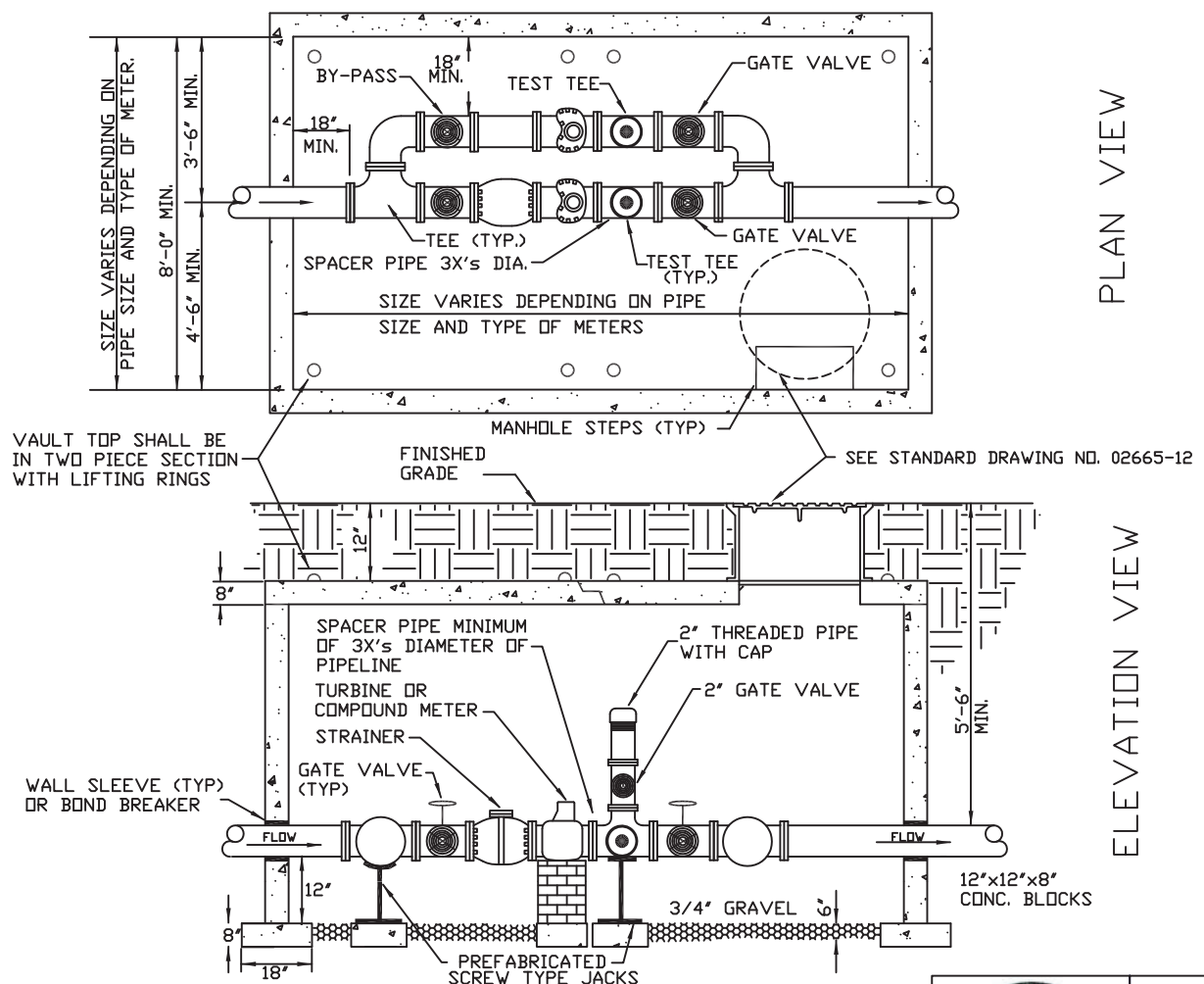


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2014

STANDARD
DRAWING NO.
02665-17

NOTES:

1. IF SURFACE IS NOT TO FINISHED GRADE AT TIME OF METER PIT INSTALLATION, OWNER SHALL RAISE OR LOWER PIT LID, WHEN SURFACE IS GRADED.
2. METER VAULT SHALL BE INSPECTED BEFORE BACKFILLING TOP.
3. BYPASS SHALL BE INSTALLED UNLESS OTHERWISE SPECIFIED.
4. ALL GATE VALVES SHALL BE DUCTILE IRON AND FITTINGS SHALL BE FLANGED.
5. ALL PIPE SHALL BE DUCTILE IRON AND ALL FITTINGS SHALL BE FLANGED.
6. SPEC. CASTING INC. CI-24-R OR AL-24-R OR APPROVED EQUIVALENT.
7. SUPPORT UNDER METER MAY BE BRICK OR CONCRETE BLOCK.
8. 3/4" ROCK SHALL BE PLACED ON FLOOR OF PIT TO A DEPTH OF 6".
9. METER ON BYPASS IS AT OPTION OF B.O.P.U. SPOOL PIECE IS TO BE PROVIDED IF METER IS NOT INSTALLED.
10. SIZE OF BYPASS TO BE DETERMINED BY B.O.P.U.
11. MANHOLE STEPS TO BEGIN 2'-4" MAX. DOWN FROM COVER AND END 1'-6" MAX. UP FROM FLOOR. STEPS TO BE SPACED 12" OR 16", SEE D.S.H.A. REQUIREMENT.
12. VAULT SHALL BE CONSTRUCTED TO HS-20-44 LOAD.



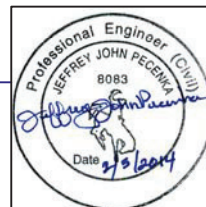
PLAN VIEW

ELEVATION VIEW



CITY OF CHEYENNE
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BOARD OF PUBLIC
UTILITIES
STANDARD DRAWING

METER SETTING FOR 3" AND
LARGER METER
WITH VALVE AND BYPASS



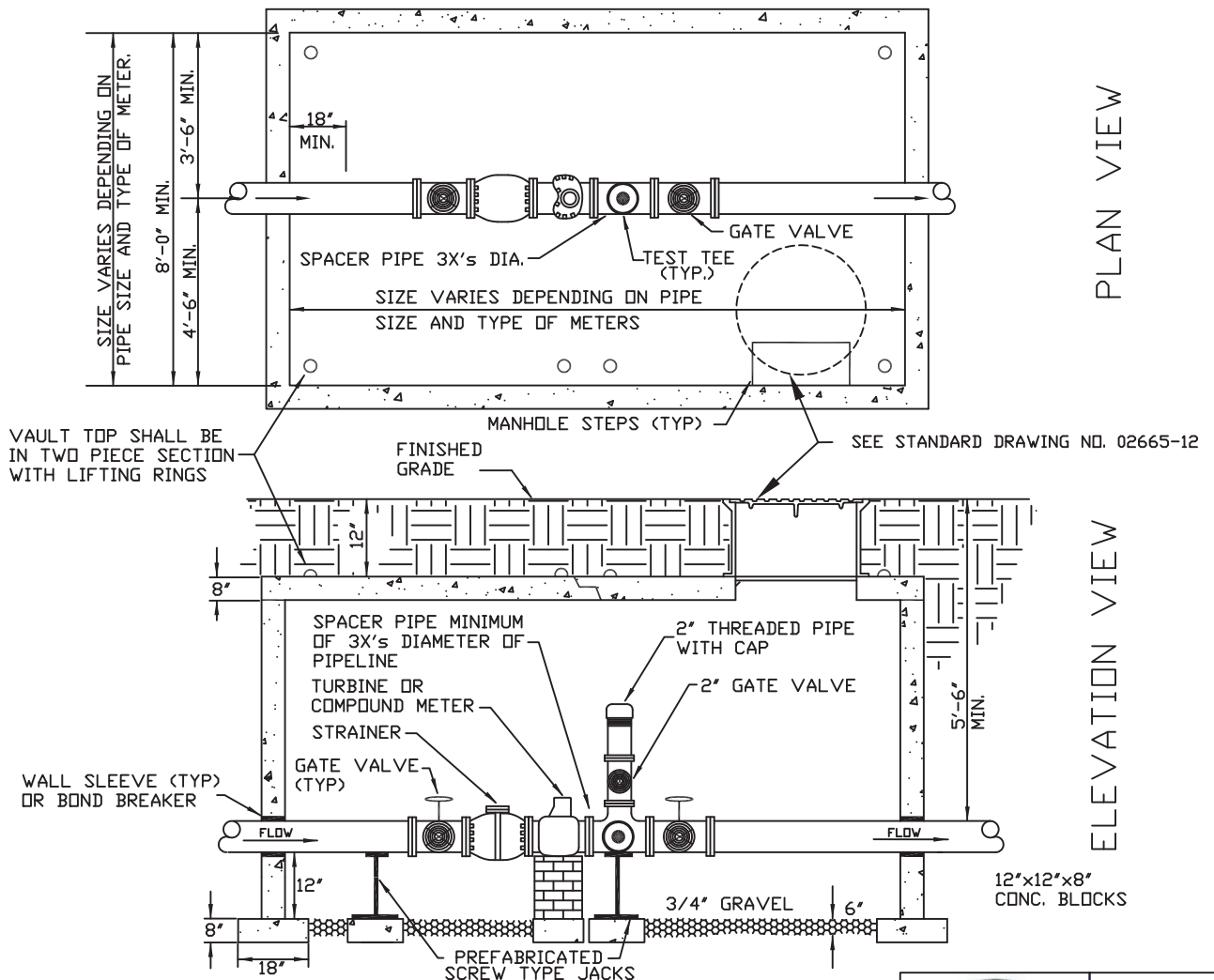
REVISED
2014

STANDARD
DRAWING NO.
02665-18

NOT TO SCALE

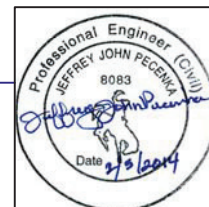
NOTES:

1. IF SURFACE IS NOT TO FINISHED GRADE AT TIME OF METER PIT INSTALLATION, OWNER MUST RAISE OR LOWER PIT LID, WHEN SURFACE IS GRADED.
2. METER VAULT SHALL BE INSPECTED BEFORE BACK FILLING TOP.
3. ALL GATE VALVES SHALL BE A NON-RISING STEM.
4. ALL PIPE SHALL BE DUCTILE IRON AND FITTINGS SHALL BE FLANGED.
5. SPEC. CASTING INC. CI-24-R, AL-24-R OR APPROVED EQUIVALENT
6. SUPPORT UNDER METER MAY BE BRICK OR CONCRETE BLOCKS.
7. 1/2" TO 3/4" ROCK SHALL BE PLACED ON FLOOR OF PIT, 6" THICK.
8. MANHOLE STEPS TO BEGIN 2'-4" MAX. DOWN FROM COVER AND END 1'-6" MAX. UP FROM FLOOR. STEPS TO BE SPACED 12" OR 16", SEE SPACING PER O.S.H.A. REQUIREMENT.
9. VAULT SHALL BE CONSTRUCTED TO HS-20-44 LOAD.
10. SIZE VARIES DEPENDING ON PIPE SIZE AND TYPE OF METER (SEE STANDARD DRAWING NO. 02665-20)



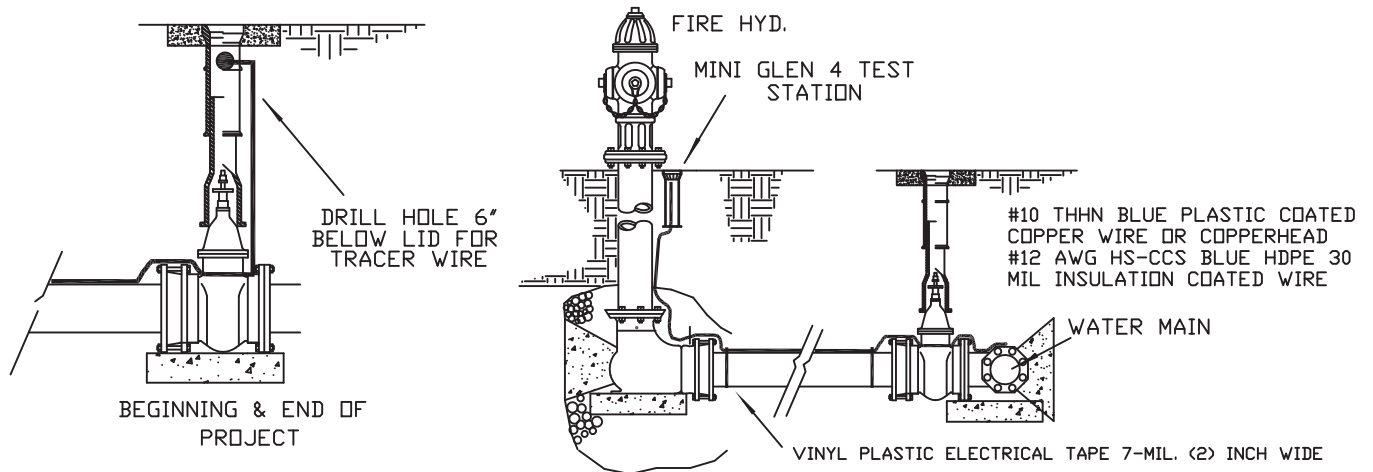
CITY OF CHEYENNE
AND
BOARD OF PUBLIC
UTILITIES
STANDARD DRAWING

IRRIGATION METER SETTING
FOR 3" AND LARGER METER



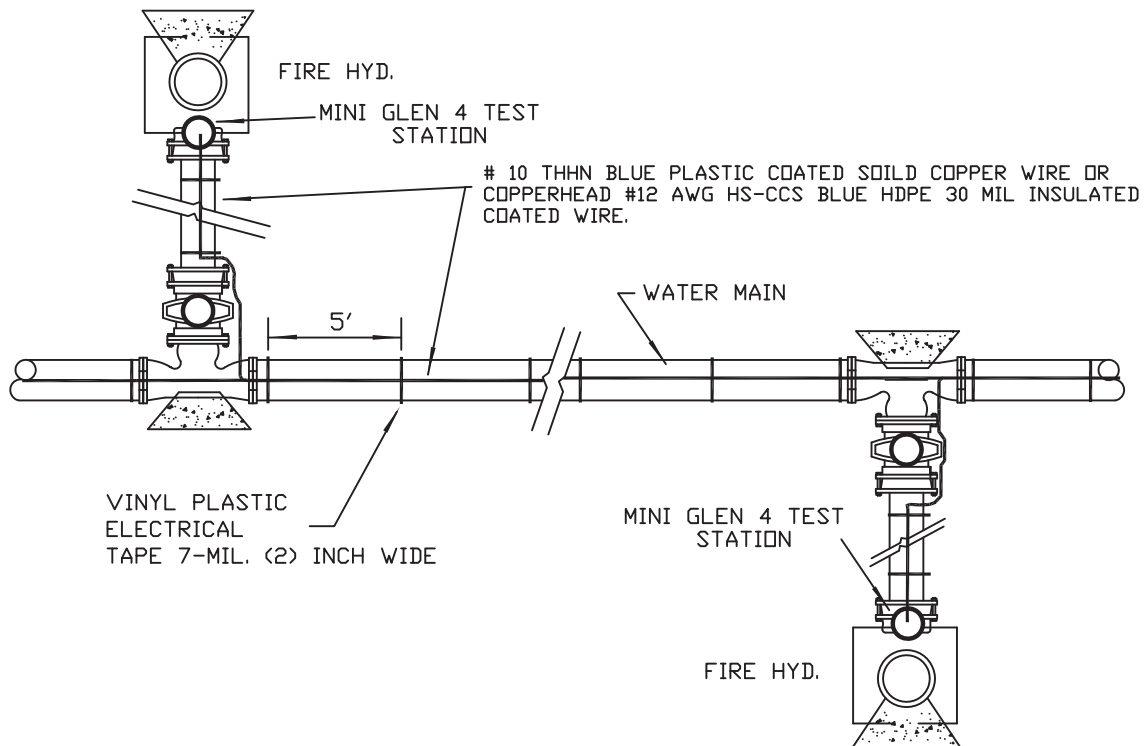
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2014

STANDARD
DRAWING NO.
02665-21



NOTES:

1. A #10 THHN BLUE PLASTIC COATED SOLID COPPER WIRE OR COPPERHEAD #12 AWG HS-CCS BLUE HDPE 30 MIL INSULATION COATED WIRE RUN CONTINUOUSLY. THE TRACER WIRE SHALL BE FASTENED SECURELY TO THE TOP OF THE PIPE WITH VINYL PLASTIC ELECTRICAL TAPE 7-MIL. (2) INCH WIDE
2. ALL SPLICES SHALL BE MADE WITH A 3M R/Y CONNECTOR, CAPABLE OF HANDLING 3-#10 WIRES.
3. AT EVERY FIRE HYDRANT LOCATION A MINI GLEN 4 STATION OR B.O.P.U. APPROVED EQUAL SHALL BE INSTALLED 1 FOOT IN FRONT OF THE HYDRANT (NOT WITHIN SIDEWALK), CONNECTED TO THE TRACER WIRE.
4. WHEN TRACER WIRE IS RUN UP THE VALVE BOX AT THE END AND BEGINNING OF THE PROJECT, RUN THE TRACER WIRE UP THE OUTSIDE OF THE VALVE BOX AND DRILL A HOLE 6 INCHES BELOW THE TOP OF THE VALVE BOX LID AND COIL 18 INCHES OF THE WIRE IN THE VALVE BOX TOP.



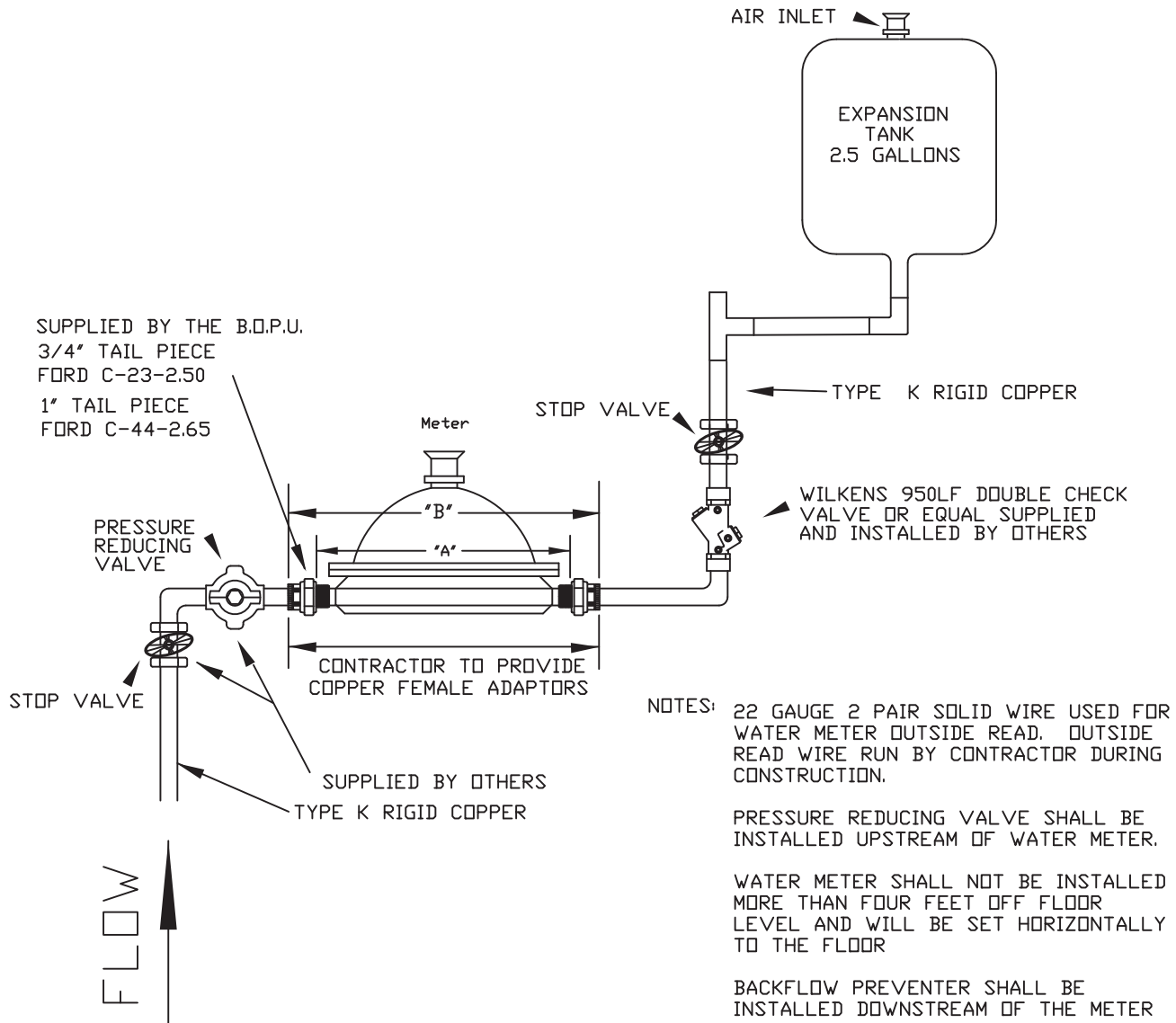
CITY OF CHEYENNE
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STANDARD DRAWING

WATER MAIN TRACER WIRE



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STANDARD
DRAWING NO.
02665-22



WARNING!

INSTALLATION OF A BACK FLOW PREVENTER REQUIRES INSTALLATION OF AN EXPANSION TANK IN ACCORDANCE WITH THE MANUFACTURER'S SPECIFICATIONS.

NO CONNECTIONS TO THE SERVICE SHALL BE PERMITTED UPSTREAM OF THE WATER METER.

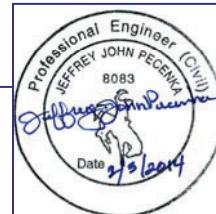
WATER METER SET CLEARANCES:
FROM BACK: 6 INCHES;
FROM SIDES: 2 FEET;
FROM FRONT: 2 FEET.

METER SIZE	"A"	"B"
3/4"	9"	13-3/4"
1"	11"	15-3/4"



CITY OF CHEYENNE
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STANDARD DRAWING

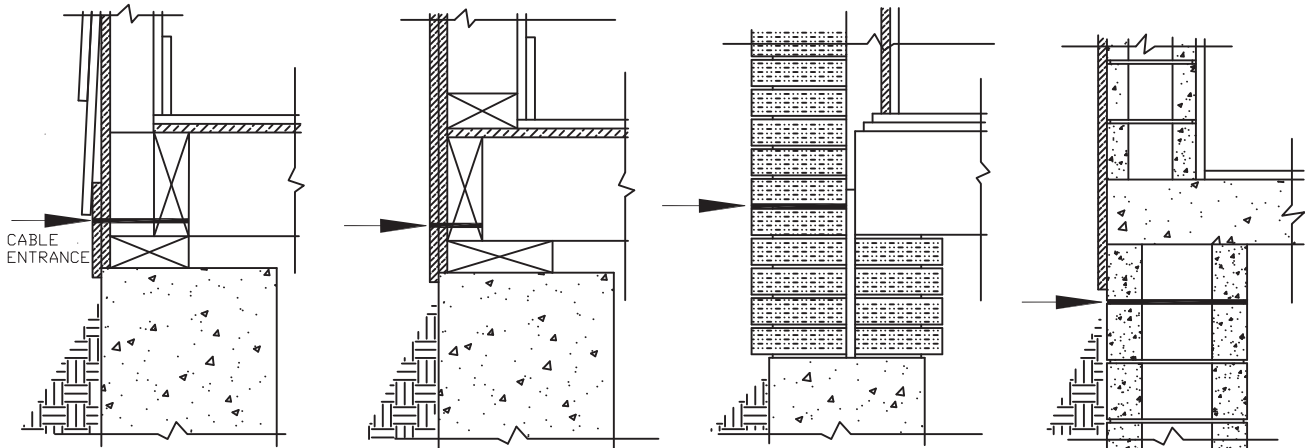
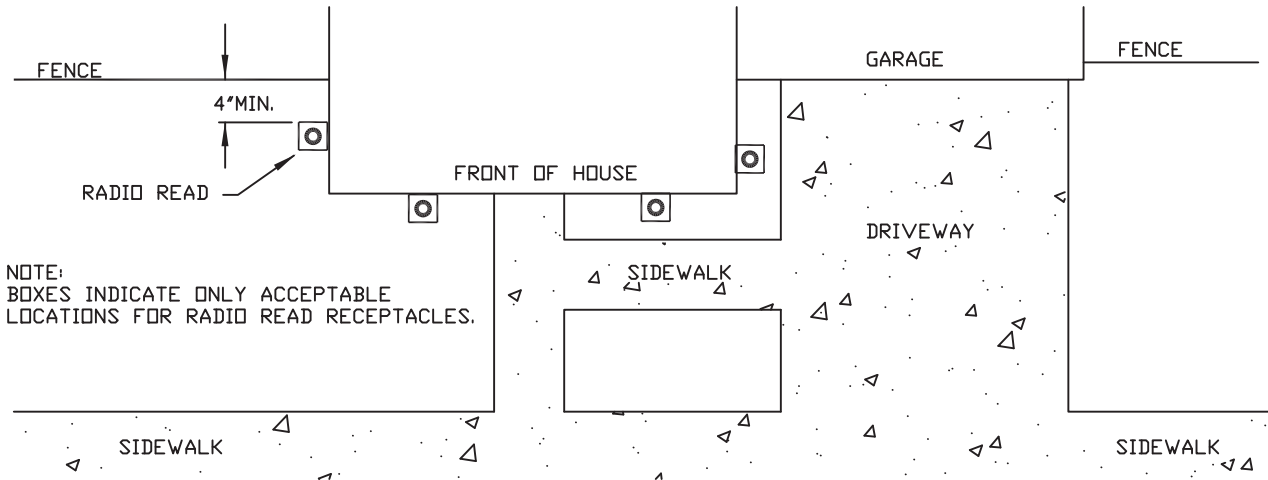
INSIDE INSTALLATION
DETAIL FOR 3/4" AND 1"
WATER METER



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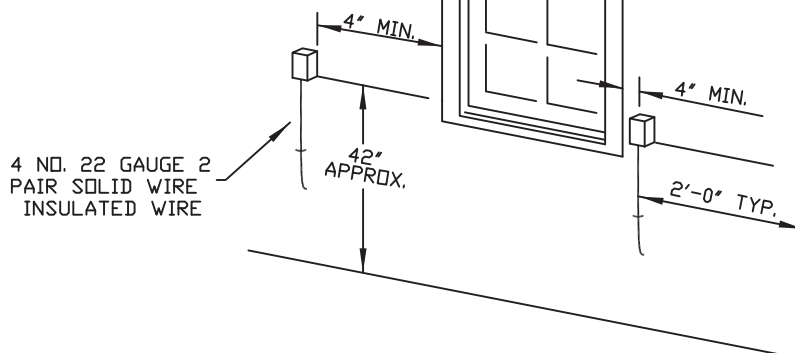
STANDARD
DRAWING NO.
02665-24

BOXES INDICATE ONLY ACCEPTABLE LOCATIONS FOR RADIO READ RECEPTACLE



SUGGESTED LOCATIONS FOR CABLE ENTRANCE HOLES IN FOUNDATIONS

NOTE:
READOUT WIRE IS REQUIRED TO BE
INSTALLED BY THE CONTRACTOR
DURING CONSTRUCTION.



NOTE:

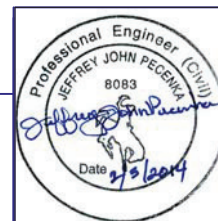
- 4" MINIMUM SPACE FROM OUTSIDE READ TO OBSTRUCTIONS
- OUTSIDE READ WIRE IS NO. 22 GAUGE, TWO PAIR, SOLID WIRE
- ALL READOUT RECEPTACLES WILL BE ACCESSIBLE TO METER READERS.
- ALL READ OUT RECEPTACLES SHALL NOT BE MORE THAN TWO FEET FROM FRONT CORNER.

EXAMPLE OF OUTSIDE RADIO READ RECEPTACLE LOCATION OUTSIDE OF HOUSE.



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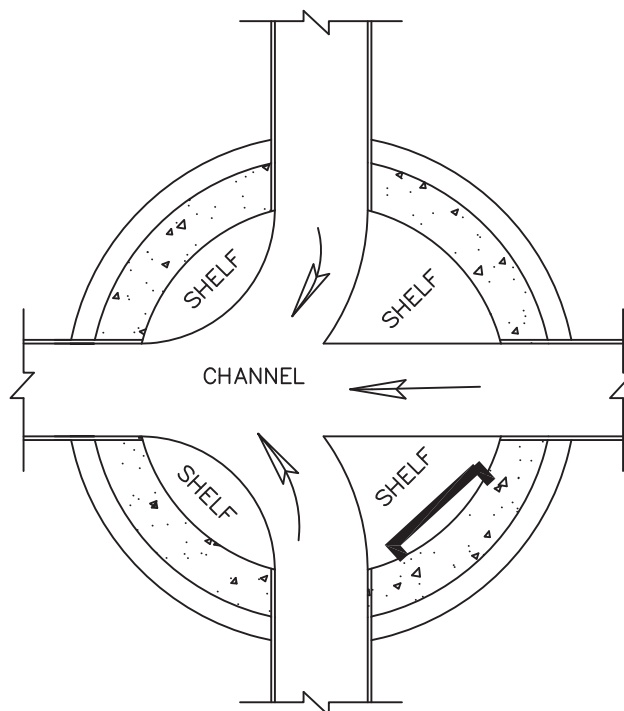
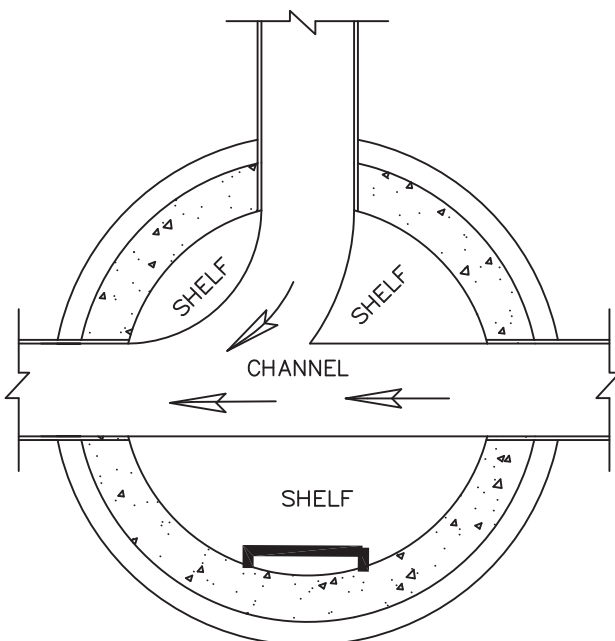
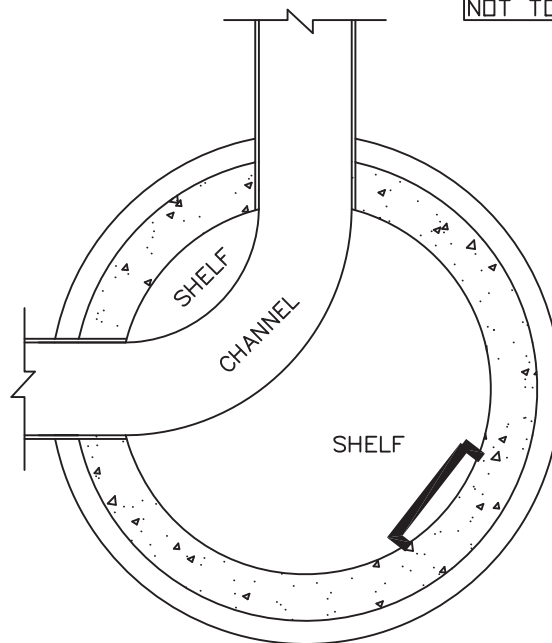
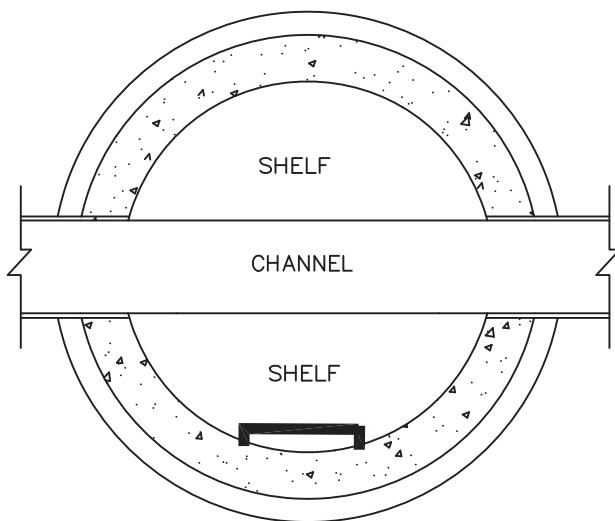
OUTSIDE RADIO READ
RECEPTACLE INSTALLATION
FOR INSIDE WATER METER



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2014

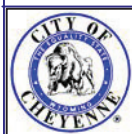
STANDARD
DRAWING NO.
02665-25

NOT TO SCALE



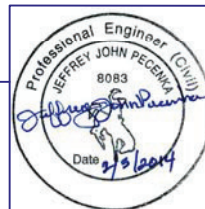
NOTES:

1. SLOPE ALL SHELVES TO CHANNEL AT 1" TO 2" PER FOOT.
2. SEE PLAN AND PROFILE SHEETS FOR SLOPE OF CHANNEL.
3. DROP ACROSS INVERT SHALL BE GREATER THAN OR EQUAL TO SLOPE OF ADJACENT SEWER, BUT NOT LESS THAN 0.1 FOOT.
4. SHELVES SHALL BE GIVEN A ROUGH BRUSHED FINISH WHEN PLACED.
5.  LOCATION FOR STEPS.
6. ALL TROUGHS SHALL BE COATED WITH A TYPE #3 EPOXY.



CITY OF CHEYENNE
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UTILITIES
STANDARD DRAWING

TYPICAL MANHOLE CHANNEL
DETAILS

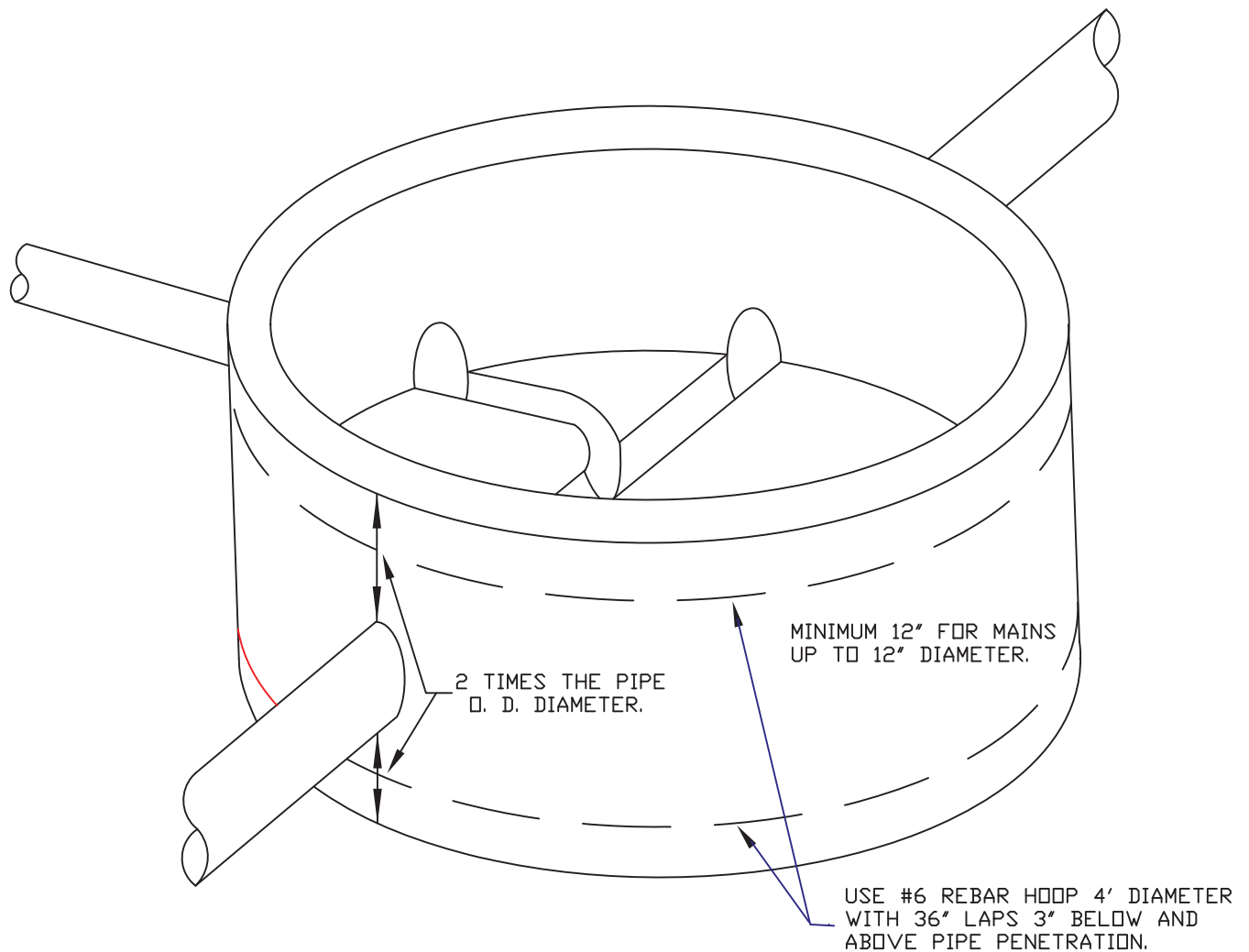


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2014

STANDARD
DRAWING NO.
02700-05

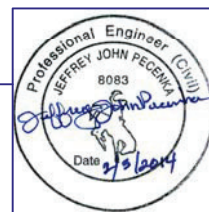
NOTES:

1. Concrete shall be four thousand (4,000) PSI meeting BOPU's concrete requirements.
2. Concrete shall be allowed to cure for five (5) day before setting M.H. sections.
3. No concrete blocks allowed.
4. Use one (1) inch to four (4) inch rock in unstable conditions.
5. Use number six (#6) rebar hoop four (4) foot diameter with thirty-six (36) inch laps three (3) inches above and below pipe penetration.
6. Use number five (#5) rebar twelve (12) inch on center below pipe penetration.
7. Concrete shall be poured two (2) times the diameter of the pipe below and above the pipe.
8. A rubber water stop material shall be around all pipe penetrations into base.
9. Double Conseal or Ramneck used when setting barrel section on to concrete base.
10. Poured in placed bases not allowed on new construction.
11. Precast and poured in place base trough's shall have an Epoxy Type 3 lining



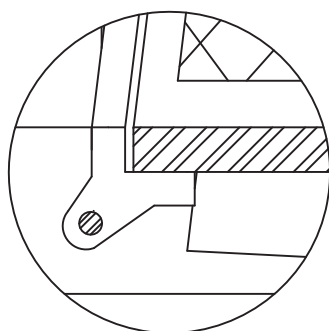
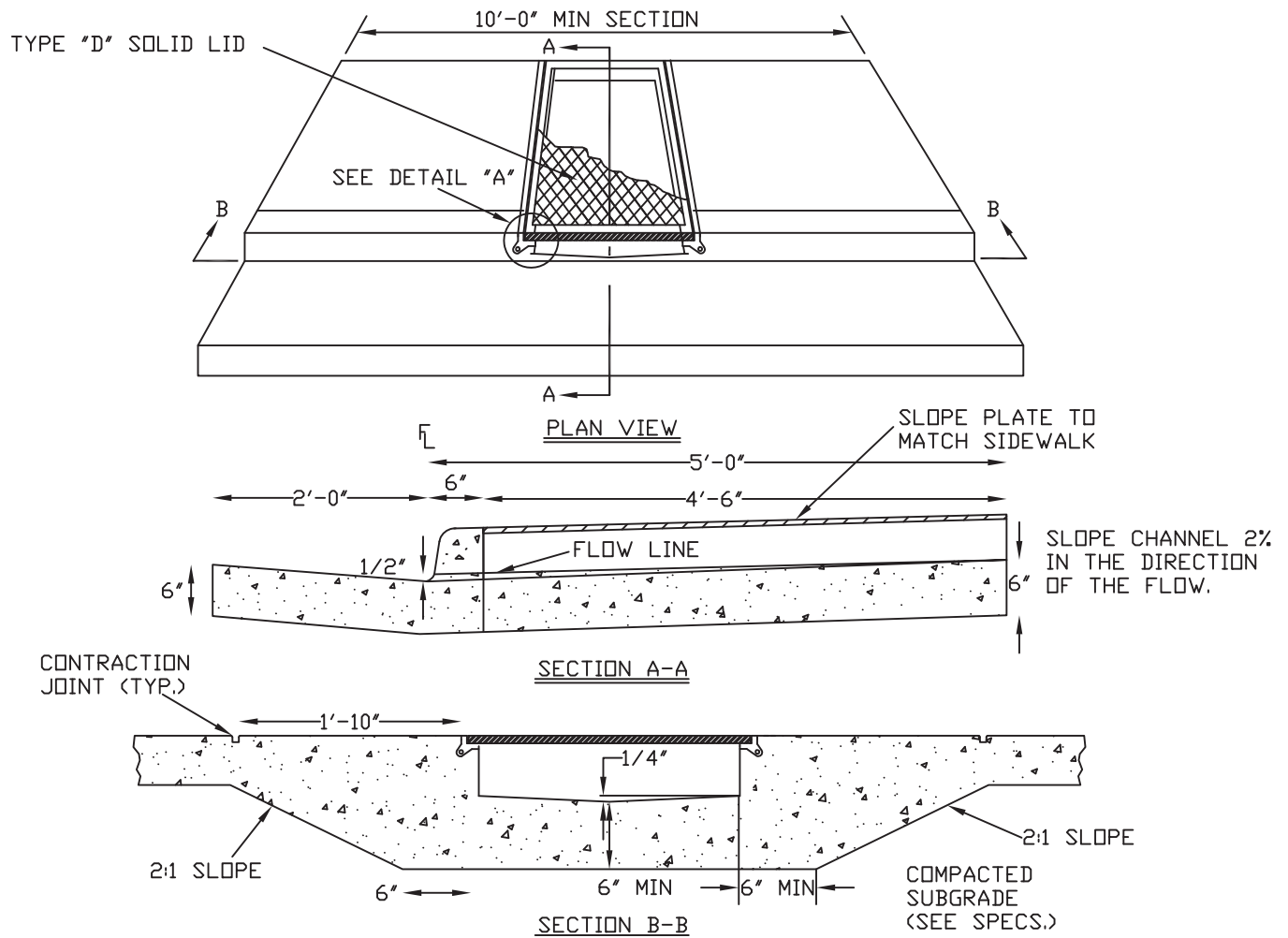
CITY OF CHEYENNE
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STANDARD DRAWING

POURED IN PLACE BASES
FOR 12" AND SMALLER SEWER



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STANDARD
DRAWING NO.
02700-06



DETAIL "A"

INSERT #3 REBAR AS ANCHOR
ROD THROUGH LENGTH OF CHASE.

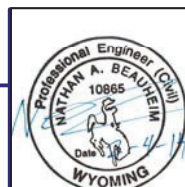
NOTES:

1. WHEN THE CURB IS SEPARATED FROM THE SIDEWALK, THE STEEL PLATE SHALL BE PLACED ON THE SIDEWALK AND THE CONCRETE CHANNEL (WITH 6" THICK WALLS ON EACH SIDE) CONTINUED INTO THE CURB AND GUTTER.
2. NEENAH R-4999HX* SERIES BOLTED TRANSVERSE DRAINAGE STRUCTURE WITH SOLID CHECKERED TYPE "D" GRATE OR EQUAL SHALL BE USED.
3. SIZE OF DRAINAGE OPENING SHALL BE CALLED OUT ON THE CIVIL CONSTRUCTION PLANS.



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SIDEWALK CHASE



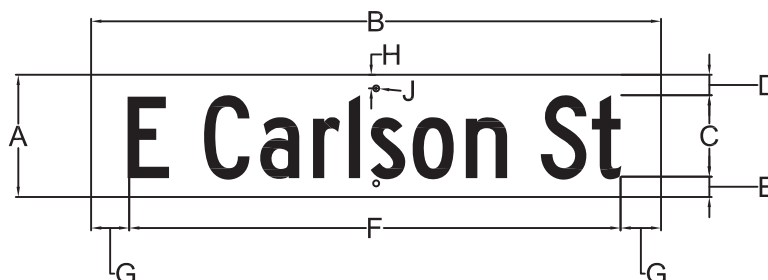
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STANDARD
DRAWING NO.
02725-04

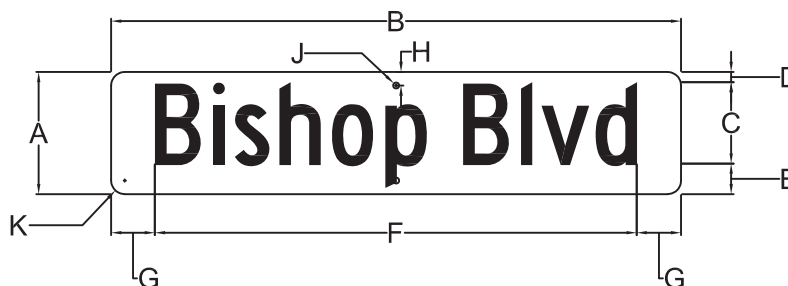
Dimensions (inches)										
	A	B	C	D	E	F	G	H	J	K
Type I	6	(1)	4C*	1	1	(2)	1.5 Min	1 (3)	$D=\frac{7}{16}$ (3)	R=1 (4)
Type I with Descenders	6	(1)	4C*	0.5	1.5	(2)	1.5 Min	1 (3)	$D=\frac{7}{16}$ (3)	R=1 (4)
Type II	9	(1)	6C**	1.5	1.5	(2)	2 Min	1 (3)	$D=\frac{7}{16}$ (3)	R=1 (4)
Type II with Descenders	9	(1)	6C**	0.75	2.25	(2)	2 Min	1 (3)	$D=\frac{7}{16}$ (3)	R=1 (4)

* Type I upper case 4" min., and lower case 3" min.

** Type II upper case 6" minimum, and lower case 4 1/2" minimum



Types I & II



Types I & II with Descenders

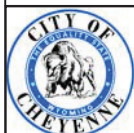
Notes

- (1) - As needed; 24" Minimum
- (2) - As needed
- (3) - Typ. of 2
- (4) - Optional

Type I Street Name Signs used at the intersection of two local roads speed limits of 25 mph or less.

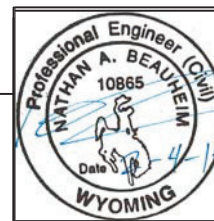
Type II Street Name Signs used at all intersections on collectors and arterials with posted speed limits of 40 mph or less or 2 lane road any speed.

COLORS: LEGEND -WHITE (RETROREFLECTIVE)
BACKGROUND -GREEN (RETROREFLECTIVE)



CITY OF CHEYENNE
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STANDARD DRAWING

TYPE I AND TYPE II
STREET NAME SIGN DETAILS

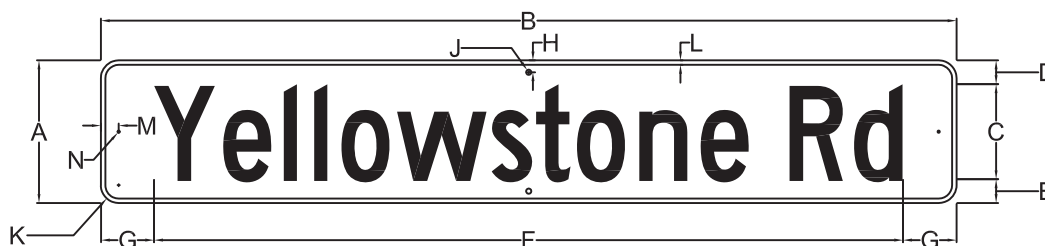


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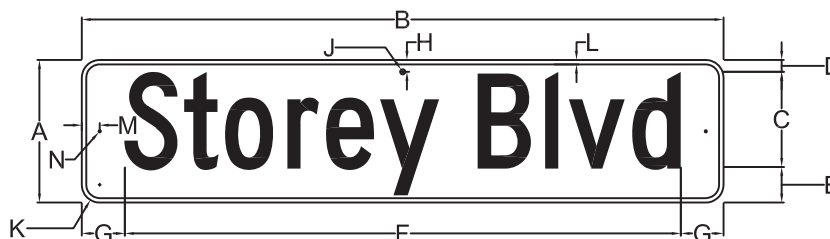
STANDARD
DRAWING NO.
02806-01

Dimensions (inches)													
	A	B	C	D	E	F	G	H	J	K	L	M	N
Type III	12	(1)	8C*	2	2	(2)	2 Min.	1 (3)	D=7/16 (3)	R=1.5	3/8	1.5 (3)	D=3/16 (3)
Type III with descenders	12	(1)	8C*	1	3	(2)	2 Min.	1 (3)	D=7/16 (3)	R=1.5	3/8	1.5 (3)	D=3/16 (3)

* Type III lower case 6" minimum



Type III



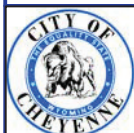
Type III with
Descenders

Notes

- (1) - As needed; 24" Minimum
- (2) - As needed
- (3) - Typ. of 2

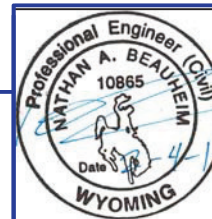
Type III Street Name Signs used at all intersections on streets with posted speed limits of more than 40 mph.

COLORS: LEGEND AND BORDER -WHITE (RETROREFLECTIVE)
BACKGROUND -GREEN (RETROREFLECTIVE)



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TYPE III
STREET NAME SIGN DETAILS



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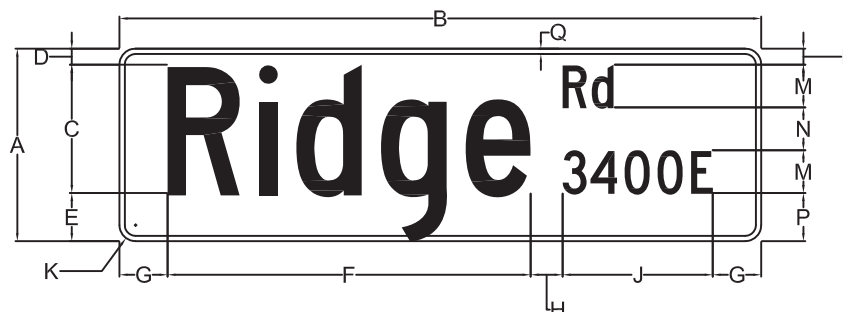
Dimensions (inches)															
	A	B	C	D	E	F	G	H	J	K	L	M	N	P	Q
Type IV	18	(1)	12C*	3	3	(2)	3 Min.	3	(2)	R=1.5	3	3C**	4	3	0.5
Type IV with descenders	18	(1)	12C*	1.5	4.5	(2)	3 Min.	3	(2)	R=1.5	1.5	3C**	4	4.5	0.5

* Type IV lower case 9" minimum

** Type IV lower case 2.25" minimum



Type IV

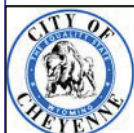
Type IV with
Descenders

Notes

- (1) - As needed; 36" Minimum
(2) - As needed

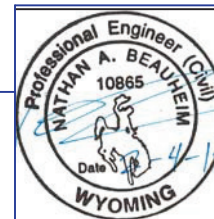
Type IV Street Name Signs to be used at all signalized intersections.

COLORS: LEGEND AND BORDER -WHITE (RETROREFLECTIVE)
BACKGROUND -GREEN (RETROREFLECTIVE)



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TYPE IV
STREET NAME SIGN DETAILS



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EXPANSION JOINTS SHALL BE
CONSTRUCTED AT THESE LOCATIONS

SLOPE NOT TO
EXCEED 1:10

REMOVE & REPLACE AS
NECESSARY TO MEET
MAX 12:1 SLOPE

SLOPE EARTH TO MEET
RAMP ELEVATIONS

5' MIN
NEW SIDEWALK ELEVATION
OLD SIDEWALK ELEVATION
SECTION A-A

C

C

SIDEWALK CURB TO
REDUCE LANDSCAPE
COST

VIEW PLAN

MAINTAIN SIDEWALK
WIDTH BETWEEN CURBS

DETECTABLE WARNING PLATES
PAVEMENT CURB & GUTTER
RAMP
SLOPE OF RAMP
MAX. 12:1
SIDEWALK
4"
THICKENED EDGE
SECTION C-C

SECTION B-B

PAY LIMITS
OF SIDEWALK

5' MIN

PERSPECTIVE

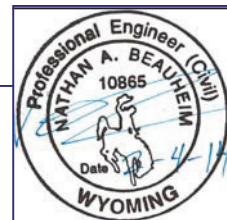
NOTES:

1. MATERIAL COMPACTION SHALL CONFORM TO SECTION 02231.
2. CONTRACTION AND EXPANSION JOINTS SHALL BE AS SHOWN AND CONFORM WITH SECTIONS 03251 AND 03340.
3. NO CONCRETE SHALL BE PLACED WITHOUT A FINAL FORM AND GRADE INSPECTION BY THE CITY CONSTRUCTION DEPARTMENT.
4. CONCRETE SHALL BE 4500 PSI WITH FIBER REINFORCEMENT AND CONFORM WITH SECTION 03304.
5. ALL LANDINGS SHALL BE 5' X 5'.
6. ENSURE ALL PAVEMENT SURFACES ARE SMOOTHLY TAPERED AND FLUSH AT CURB RAMPS TO FACILITATE WHEELCHAIR MOVEMENTS
7. DETECTABLE WARNINGS SHALL BE EAST JORDON IRON WORKS 7005 SERIES NATURAL FINISH (7006 SERIES ON RADIUS TYPE), NEENAH 4984 SERIES DETECTABLE WARNING PLATES OR APPROVED EQUAL.
8. DETECTABLE WARNING PLATES SHALL BE PLACED AT BOTTOM OF RAMP AND EXTEND two (2') FEET BACK FROM EDGE AND A MINIMUM LENGTH OF 5 FEET (TO MATCH WIDTH OF RAMP).



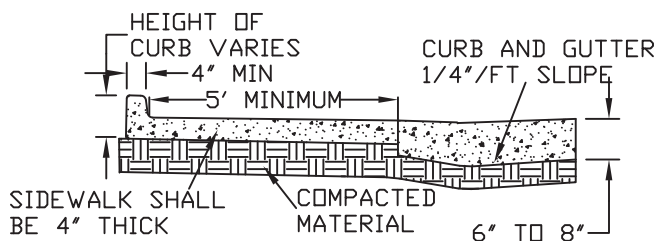
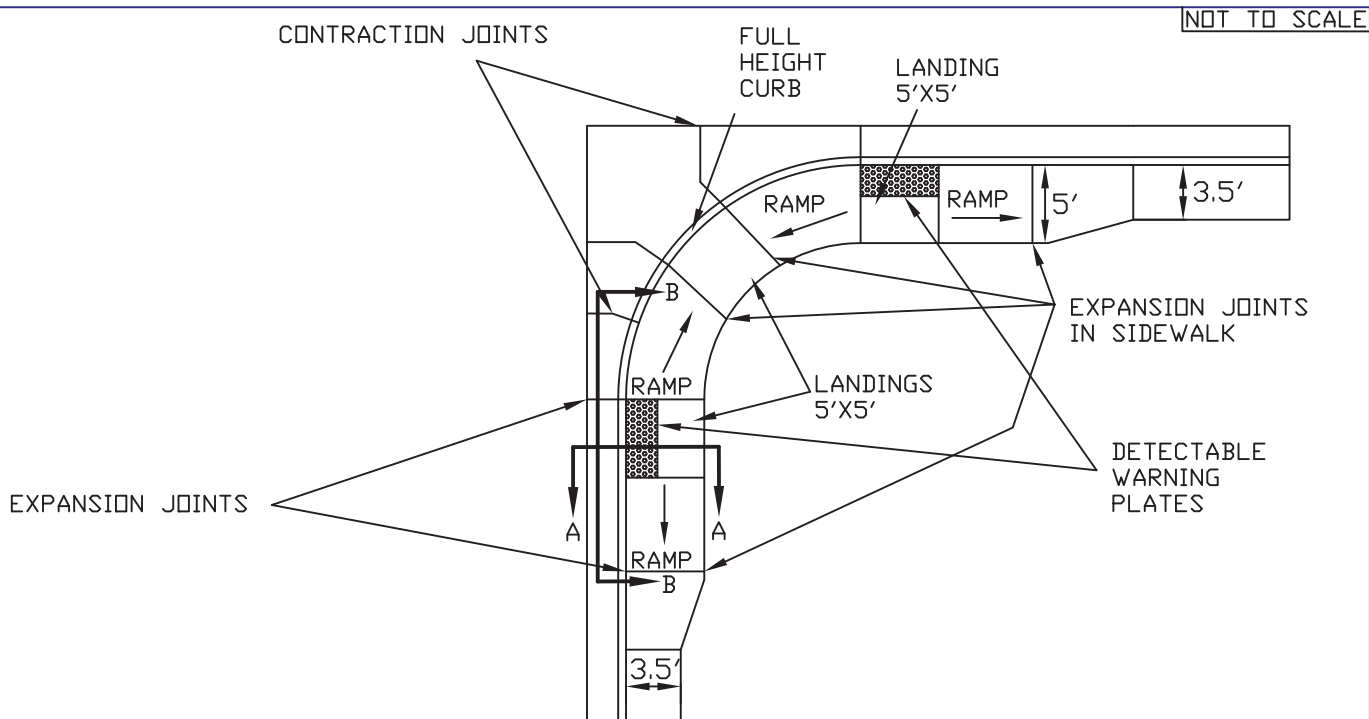
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WHEELCHAIR RAMP TYPE "A" WITH DETECTABLE WARNING PLATES

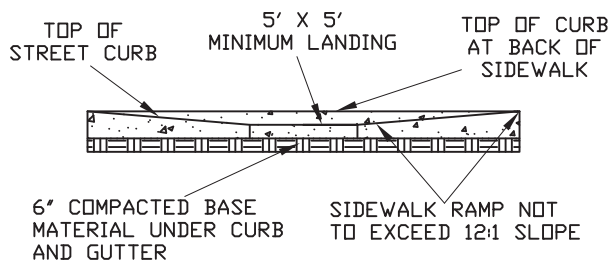


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03340-02



SECTION A-A



SECTION B-B

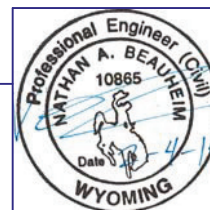
NOTES:

1. MATERIAL COMPACTION SHALL CONFORM TO SECTION 02231.
2. CONTRACTION AND EXPANSION JOINTS SHALL BE AS SHOWN AND CONFORM WITH SECTIONS 03251 AND 03340.
3. NO CONCRETE SHALL BE PLACED WITHOUT A FINAL FORM AND GRADE INSPECTION BY THE CITY CONSTRUCTION DEPARTMENT.
4. CONCRETE SHALL BE 4500 PSI WITH FIBER REINFORCEMENT AND CONFORM WITH SECTION 03405.
5. ALL LANDINGS SHALL BE 5' X 5'.
6. ENSURE ALL PAVEMENT SURFACES ARE SMOOTHLY TAPERED AND FLUSH AT CURB RAMPS TO FACILITATE WHEELCHAIR MOVEMENTS.
7. DETECTABLE WARNINGS SHALL BE EAST JORDON IRON WORKS 7005 SERIES NATURAL FINISH (7006 SERIES ON RADIUS TYPE) DETECTABLE WARNING PLATES OR APPROVED EQUAL.
8. DETECTABLE WARNING PLATES SHALL BE PLACED AT BOTTOM OF RAMP AND EXTEND 2 FEET BACK FROM EDGE AND A MINIMUM LENGTH OF 5 FEET (TO MATCH WIDTH OF RAMP).



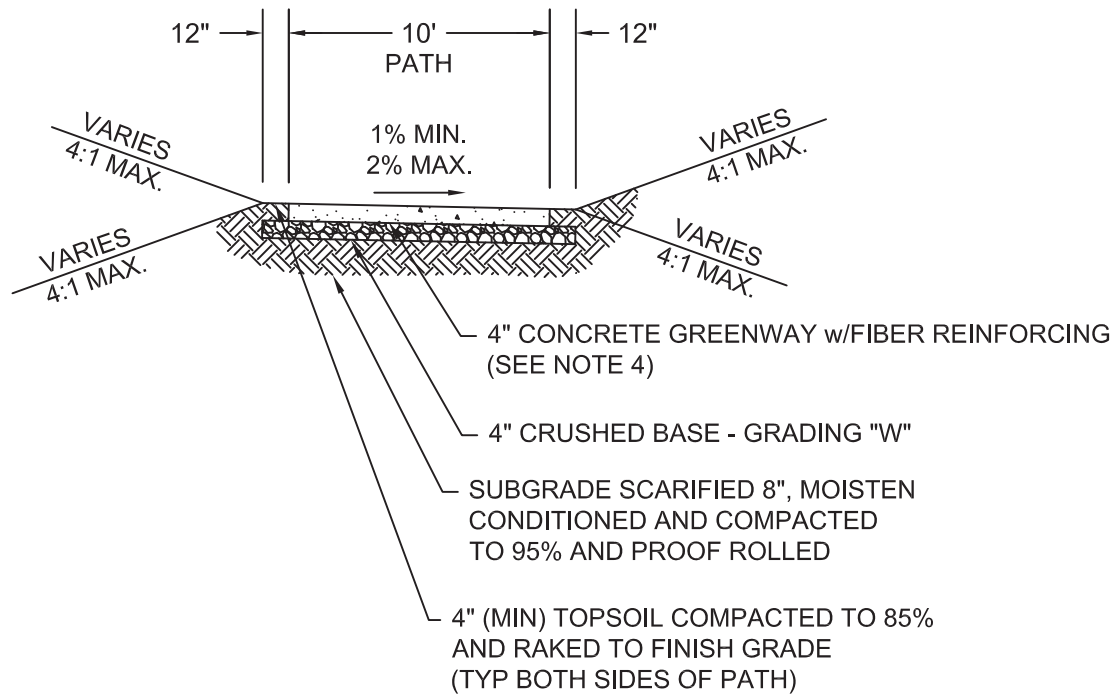
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WHEEL CHAIR RAMP
TYPE "B", WITH
DETECTABLE WARNING PLATES



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DRAWING NO.
03340-03



TYPICAL SECTION

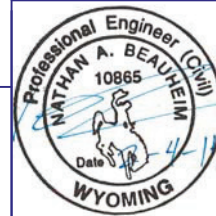
NOTES:

- 1.) FULL WIDTH TRANSVERSE GREENWAY JOINT SPACING SHALL BE 10'-0" (SAWCUT). 1" MIN. DEPTH.
- 2.) EXPANSION GREENWAY JOINT SPACING SHALL BE 150'. USE PREFORMED JOINT MATERIAL FULL DEPTH. GROUT AND SEAL WITH SILICONE. EXPANSION JOINT SHALL BE CONSIDERED SUBSIDIARY TO THE GREENWAY.
- 3.) ALL CONCRETE SHALL BE 4500 PSI WITH FIBER REINFORCEMENT AND CONFORM TO SECTION 03405.
- 4.) ALL GREENWAY PATH WILL BE 4" THICK EXCEPT FOR ACROSS APPROACHES. THICKNESS OF DRIVE APPROACH SHALL BE 6" FOR RESIDENTIAL AND 8" FOR COMMERCIAL DRIVE AND ALLEY APPROACHES.
- 5.) CRUSHED BASE IS TO EXTEND 12" PAST EDGE OF PATH AND COMPACTED PER SECTION 02231.
- 6.) NO CONCRETE SHALL BE PLACED WITHOUT A FINAL FORM AND GRADE INSPECTION BY THE CITY.
- 7.) MAX. LONGITUDINAL GRADE SHALL NOT EXCEED 5% PER ADA STANDARDS.



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GREENWAY TYPICAL SECTION



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